

STS624 / STS724

Generalized Linear Models (with Applications using SAS)

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Dobson AJ, Barnett AG: *An Introduction to Generalized Linear Models, 3rd Edition*.

Boca Raton, FL: CRC Press, 2008 (Chapters 1–6)

McCullagh P, Nelder JA: *Generalized Linear Models, 2nd Edition*. London: Chapman & Hall, 1989 (Section 2.6, Section 5.7, Section 6.8, Chapter 7)

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Nature and Aims of Course

Nature of the Course

- Present statistical methods for data arising from generalized linear models
- Apply methods to real data
- Use SAS software package to perform analyses
- Discuss SAS output
- Interpret results

Learning aims

- Understand the statistical *concepts*
- Understand which methods to apply when (to which data)
- Learn how to model data generated from exponential family
- Learn how to use SAS (PROC GENMOD) to fit generalized linear models
- Understand relevant output from SAS PROC GENMOD
- Interpret results of data analyses using generalized linear models

Chapter 1

Introduction

1.1 Linear Models

The conventional **linear model** can be specified in the general form

$$y_i = \mu + x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p + e_i \quad (1.1)$$

where

- y_i represents an observation of the **dependent** variable
- μ and $\beta_1, \dots, \beta_p$ are $p + 1$ unknown **parameters**
- $x_{i1}, \dots, x_{ip}$ are the **known** values of the p **independent** variables
- e_i is the **error term**

With Model (1.1) we assume that

$$E(Y_i) = \mu + x_{i1}\beta_1 + x_{i2}\beta_2 + \cdots + x_{ip}\beta_p \quad (1.2)$$

Equation (1.2) states that the expected value of Y_i can be written as a **linear** combination of the values of the independent variables. From equation (1.2) it follows that

$$E(e_i) = 0$$

If there are n observations, the model equations for the linear model can be written as

$$\begin{aligned} y_1 &= \mu + x_{11}\beta_1 + x_{12}\beta_2 + \cdots + x_{1p}\beta_p + e_1 \\ y_2 &= \mu + x_{21}\beta_1 + x_{22}\beta_2 + \cdots + x_{2p}\beta_p + e_2 \\ &\vdots \\ y_n &= \mu + x_{n1}\beta_1 + x_{n2}\beta_2 + \cdots + x_{np}\beta_p + e_n \end{aligned} \tag{1.3}$$

Often, the assumption is made that the errors e_i , $i = 1, \dots, n$ are independently and identically distributed, with

$$\text{Var}(e_i) = \sigma^2, \quad i, \dots, n$$

Linear regression, Analysis of Variance (ANOVA) and Analysis of Covariance (ANCOVA) are all examples of the application of Model (1.1)/(1.3). Common to all these models is the fact that the mean or the expectation of the response variable Y_i is modelled as a **linear function** of a number of parameters β . Furthermore, in these applications the distribution of the errors e_i (and therefore also of the observations y_i) is often assumed to be the **Normal** distribution.

1.2 Examples: Linear Models

We can illustrate several types of linear models (of the form (1.1) and (1.3)) with the following data, which are the first nine observations from a multi-centre clinical trial of two treatments, Treatment A and Treatment B, to lower blood pressure:

Table 1.1: Data from Clinical Trial

Centre	Treatment	Pre-treatment SBP	Post-treatment SBP
1	A	178	176
1	A	168	194
1	B	196	156
1	B	170	150
2	A	165	150
2	B	190	160
3	A	175	150
3	A	180	160
3	B	175	160

SBP: systolic blood pressure

We have data on Post-treatment systolic blood pressure (PostSBP) and on Pre-treatment SBP (PreSBP), from three clinical trial centres (1–3).

1.2.1 Simple linear regression

Let us assume we want to perform a simple linear regression of PostSBP against PreSBP. The simple linear regression model, written in the form (1.3) for the 9 observations from Table 1.1, would then look as follows:

$$\begin{aligned}
 176 &= \mu + 178\beta_1 + e_1 \\
 194 &= \mu + 168\beta_1 + e_2 \\
 156 &= \mu + 196\beta_1 + e_3 \\
 &\vdots \\
 160 &= \mu + 180\beta_1 + e_8 \\
 160 &= \mu + 175\beta_1 + e_9
 \end{aligned} \tag{1.4}$$

In Model (1.4), as in all simple linear regression models, we fit an intercept parameter μ and a slope parameter β_1 . The independent variable is formed by $x_{11} = 178, x_{21} =$

168, ..., $x_{91} = 175$.

The SAS PROC GENMOD code to fit the above simple linear regression model is as follows:

```
PROC GENMOD DATA=sbp;
  TITLE 'Simple Linear Regression';
  MODEL postsbp = presbp / DIST=normal;
RUN;
```

Note that PROC GENMOD would usually not be used to fit a simple linear regression model, since other SAS procedures are more suited for fitting linear models (PROC REG, PROC GLM, PROC MIXED). However, as we shall see, the conventional Normal linear model is a special case of a generalized linear model, and the SAS code here is given to illustrate how linear models (linear predictors) are specified in SAS generally.

Both the outcome variable, **postsbp** on the left hand side of the equal sign, and the predictor variable, **presbp** on the right hand side of the equal sign, are specified in the **MODEL** statement. The model specification is practically complete: the only additional specification needed (in SAS PROC GENMOD) is the error distribution, which is Normal (see the **DIST=normal** option in the **Model** statement).

The output for the above SAS code is included with Appendix 1.1.

1.2.2 Two-way Analysis of Variance

Let us now assume we would like to do a two-way analysis of variance (ANOVA) of PastSBP, fitting Treatment and Centre effects as explanatory variables. The associated linear model is sometimes written in the following form:

$$y_{ij} = \mu + c_i + t_j + e_{ij}; \quad i = 1, 2, 3; \quad j = 1, 2 \quad (1.5)$$

where c_1 , c_2 and c_3 represent the three Centre effects, and t_1 and t_2 the two treatment effects. Thus Model (1.5) uses (double) subscripts ij to denote results for the j -th subject from the i -th centre. In the general notation of Model (1.3), however, every observation is denoted separately with a single subscript i ; thus $y_1, \dots, y_4$ would denote the observations for 4 subjects from centre 1, y_5 and y_6 the observations for 2 subjects

from centre 2, etc. The β terms in the general notation (1.3) correspond then to the 3 centre effects c_1 , c_2 and c_3 , and to the two treatment effects t_1 and t_2 . In this case, the independent variables $x_{i1}, x_{i2}, \dots, x_{i5}$ are used to indicate the centre and treatment to which observation y_i belongs; for this purpose the independent variables can only take on the values 1 or 0. Thus if y_1 is the observation from centre 1 receiving treatment 1, then $x_{11} = 1$ (corresponding to β_1 , which represents the first centre effect), $x_{12} = x_{13} = 0$ (since the observation is not from centres 2 or 3), $x_{14} = 1$ (corresponding to β_4 , representing the first treatment effect), and $x_{15} = 0$.

The two-way ANOVA model, written in the form (1.3) for the 9 observations from Table 1.1, would then look as follows:

$$\begin{aligned}
 176 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + e_1 \\
 194 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + e_2 \\
 156 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 0 \cdot \beta_4 + 1 \cdot \beta_5 + e_3 \\
 &\vdots \\
 160 &= 1 \cdot \mu + 0 \cdot \beta_1 + 0 \cdot \beta_2 + 1 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + e_8 \\
 160 &= 1 \cdot \mu + 0 \cdot \beta_1 + 0 \cdot \beta_2 + 1 \cdot \beta_3 + 0 \cdot \beta_4 + 1 \cdot \beta_5 + e_9
 \end{aligned} \tag{1.6}$$

Note that Model (1.6) is in the so-called overparameterized form used by software packages such as SAS (in procedures PROC GLM, PROC MIXED and PROC GENMOD).

The SAS PROC GENMOD code to fit the above two-way ANOVA model is as follows:

```

PROC GENMOD DATA=sbp;
  TITLE 'Two-Way ANOVA';
  CLASS centre treat;
  MODEL postsbp = centre treat / DIST=normal;
RUN;

```

The output for the above SAS code is included with Appendix 1.2.

The **CLASS** statement specifies that centre and treatment are categorical (Class) variables. Both categorical variables, centre and treatment, are fitted in the **MODEL**

statement. If we wanted to fit the interaction between centre and treatment as well, the MODEL statement would look as follows:

```
MODEL postsbp = centre treat centre*treat / DIST=normal;
```

1.2.3 Analysis of Covariance

Building on Model (1.6) we now would like to fit Pre-treatment SBP as a covariate, in addition to the Treatment and Centre effects. We thus have an analysis of covariance (ANCOVA) model with PreSBP as covariate and Treatment and Centre as categorical effects. Effectively, we fit another parameter β_6 (representing the covariate effect) with corresponding independent variable $x_{16} = 178, x_{26} = 168, \dots, x_{96} = 175$, similar to Model (1.6).

The ANCOVA model, written in the form (1.3) for the 9 observations from Table 1.1, would then look as follows:

$$\begin{aligned}
 176 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + 178 \cdot \beta_6 + e_1 \\
 194 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + 168 \cdot \beta_6 + e_2 \\
 156 &= 1 \cdot \mu + 1 \cdot \beta_1 + 0 \cdot \beta_2 + 0 \cdot \beta_3 + 0 \cdot \beta_4 + 1 \cdot \beta_5 + 196 \cdot \beta_6 + e_3 \\
 &\vdots \\
 160 &= 1 \cdot \mu + 0 \cdot \beta_1 + 0 \cdot \beta_2 + 1 \cdot \beta_3 + 1 \cdot \beta_4 + 0 \cdot \beta_5 + 180 \cdot \beta_6 + e_8 \\
 160 &= 1 \cdot \mu + 0 \cdot \beta_1 + 0 \cdot \beta_2 + 1 \cdot \beta_3 + 0 \cdot \beta_4 + 1 \cdot \beta_5 + 175 \cdot \beta_6 + e_9
 \end{aligned} \tag{1.7}$$

The SAS PROC GENMOD code to fit the above ANCOVA model is as follows:

```
PROC GENMOD DATA=sbp;
  TITLE 'Analysis of Covariance (ANCOVA)';
  CLASS centre treat;
  MODEL postsbp = presbp centre treat / DIST=normal;
RUN;
```

The output for the above SAS code is included with Appendix 1.3.

In general, covariates are fitted by simply listing the corresponding variable (here:

presbp) on the right hand side of the equality sign in the `MODEL` statement. Categorical variables are fitted in the same way, but in order to specify which variables are categorical the `MODEL` statement must be preceded by a `CLASS` statement.

1.3 Matrix Notation for Linear Models

As we have seen in Section (1.1), all linear models can be written in the general form (1.3). These equations can be written more concisely in matrix notation as

$$y = X\beta + e, \quad \text{Cov}(Y) = \sigma^2 I_n \quad (1.8)$$

where

$$\begin{aligned} y &= (y_1, \dots, y_n)' &&= \text{vector of observations,} \\ \beta &= (\mu = \beta_0, \beta_1, \dots, \beta_p)' &&= \text{model parameters,} \\ e &= (e_1, \dots, e_n)' &&= \text{residuals,} \\ \sigma^2 &&&= \text{residual variance,} \\ I_n &&&= n \times n \text{ identity matrix.} \end{aligned}$$

The parameters in β may be associated with several different variables. In the example of Section 1.2.3 above, the parameters $\beta = (\beta_0, \beta_1, \dots, \beta_7)'$ of the ANCOVA model are respectively associated with the intercept, with 3 centre effects, with 2 treatment effects, and with a covariate effect. Both the centre and treatment effects are categorical variables, and one can refer to such effects as categorical effects; they are also sometimes referred to as factor effects. More generally, categorical effects are those where observations belong to one of several classes. (In SAS procedures like GLM and GENMOD, the `CLASS` statement tells SAS which of the variables in the `MODEL` statement are categorical.)

The vector of parameters β may contain one or more covariate effects (such as baseline measurement: PreSBP in the example). These covariate effects relate to variables which are measured on a quantitative scale. While several parameters are required to model a categorical effect (namely one parameter for each category), only one parameter is used to model a covariate effect.

The matrix

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & & & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix} \quad (1.9)$$

is known as the **design matrix** and has dimension $n \times (p + 1)$, namely n rows (for n observations) and $p + 1$ columns (for $p + 1$ parameters). For each observation, the matrix X specifies the fixed effects corresponding to each parameter. For categorical effects the values of 0 and 1 are used to denote the presence or absence of effect categories, and for covariate effects the covariate values themselves are used in X . Usually, the first column of X is a vector of 1's, corresponding the model intercept which is fitted in almost all linear models.

Note: Because of the special place of the intercept in linear models, above we have specified the parameter vector β as having dimension $p + 1$, namely containing μ , plus p parameters $\beta_1, \dots, \beta_p$. Similarly, the design matrix has $p + 1$ columns. However, it is often convenient to include the intercept among the p model parameters, corresponding to a design matrix with p columns.

The design matrix X and parameter vector β are used to model the mean (or expected value) of Y . In matrix notation we have

$$\mu = E(Y) = X\beta \quad (1.10)$$

where $\mu = (\mu_1, \dots, \mu_n)'$. For each observation y_i ,

$$\mu_i = E(Y_i) = x'_i\beta \quad (1.11)$$

where x'_i is the i -th row of the design matrix (see (1.9)).

1.4 Examples: Matrix Notation for Linear Models

1.4.1 Simple linear regression

Model (1.4) in matrix notation is written as follows:

$$y = X\beta + e$$

$$\Leftrightarrow \begin{pmatrix} 176 \\ 194 \\ 156 \\ 150 \\ 150 \\ 160 \\ 150 \\ 160 \\ 160 \end{pmatrix} = \begin{pmatrix} 1 & 178 \\ 1 & 168 \\ 1 & 196 \\ 1 & 170 \\ 1 & 165 \\ 1 & 190 \\ 1 & 175 \\ 1 & 180 \\ 1 & 175 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \end{pmatrix} \quad (1.12)$$

where the 2 columns of the design matrix correspond to the 2 parameters in β , namely

$\mu = \beta_0 =$ intercept

$\beta_1 =$ parameter (slope) representing covariate effect (pre-treatment SBP)

1.4.2 Two-way Analysis of Variance

The two-way ANOVA Model (1.6) in matrix notation is written as follows:

$$y = X\beta + e$$

$$\Leftrightarrow \begin{pmatrix} 176 \\ 194 \\ 156 \\ 150 \\ 150 \\ 160 \\ 150 \\ 160 \\ 160 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \end{pmatrix} \quad (1.13)$$

where the 6 columns of the design matrix correspond to the 6 parameters in β , namely

$\mu = \beta_0 =$ intercept

$\beta_1, \beta_2, \beta_3 =$ parameters representing centre effects

$\beta_4, \beta_5 =$ parameters representing treatment effects

1.4.3 Analysis of Covariance

Finally, the ANCOVA Model (1.7) in matrix notation is written as follows:

$$\begin{aligned}
 y &= X\beta + e \\
 \Leftrightarrow \begin{pmatrix} 176 \\ 194 \\ 156 \\ 150 \\ 150 \\ 160 \\ 150 \\ 160 \\ 160 \end{pmatrix} &= \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 178 \\ 1 & 1 & 0 & 0 & 1 & 0 & 168 \\ 1 & 1 & 0 & 0 & 0 & 1 & 196 \\ 1 & 1 & 0 & 0 & 0 & 1 & 170 \\ 1 & 0 & 1 & 0 & 1 & 0 & 165 \\ 1 & 0 & 1 & 0 & 0 & 1 & 190 \\ 1 & 0 & 0 & 1 & 1 & 0 & 175 \\ 1 & 0 & 0 & 1 & 1 & 0 & 180 \\ 1 & 0 & 0 & 1 & 0 & 1 & 175 \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \\ \beta_5 \\ \beta_6 \end{pmatrix} + \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \end{pmatrix} \quad (1.14)
 \end{aligned}$$

where the 7 columns of the design matrix correspond to the 7 parameters in β , namely

$\mu = \beta_0$ = intercept

$\beta_1, \beta_2, \beta_3$ = parameters representing centre effects

β_4, β_5 = parameters representing treatment effects

β_6 = parameter (slope) representing covariate effect (pre-treatment SBP)

Chapter 2

Generalizing the Linear Model

2.1 Introduction

Advances in statistical theory and in computer software allow us to use methods analogous to those developed for linear models in the following more general situations:

1. Response variables have distributions other than the Normal distribution – response variables may even be categorical rather than continuous.
2. The relationship between the response variable and the explanatory variables need not be of the simple linear form as in (1.1) or (1.8).

One of the advances has been the recognition that many of the “convenient” properties of the Normal distribution are shared by a wider class of distributions called the **exponential family** of distributions. These distributions and some of their properties are discussed in the next section.

A second advance is the extension of the numerical methods to estimate the parameters β from the linear model (1.8) to the situation where there is a non-linear function relating $E(y_i) = \mu_i$ to the linear predictor $x_i'\beta$. That is, we now assume, more generally than in (1.2), that

$$g(\mu_i) = x_i'\beta \tag{2.1}$$

The function $g(\cdot)$, which may be non-linear, is called the **link function**.

This chapter introduces the exponential family of distributions and defines **generalized linear models**. Methods for parameter estimation and hypothesis testing in generalized linear models will be presented in Chapters 3 and 4, respectively.

2.2 Exponential Family of Distributions

Consider a single random variable Y whose distribution depends on the single parameter θ . The distribution belongs to the exponential family of distributions if its probability density function (pdf; for continuous distributions) or probability function (pf; for discrete distributions) can be written in the form

$$f(y; \theta) = \exp[a(y)b(\theta) + c(\theta) + d(y)] \quad (2.2)$$

If $a(y) = y$ the distribution is said to be in **canonical form** (standard form), and $b(\theta)$ is called the **natural parameter** of the distribution.

If there are other parameters, in addition to the parameter of interest, θ , they are said to be **nuisance parameters** forming parts of the functions $a(\cdot)$, $b(\cdot)$, $c(\cdot)$ and $d(\cdot)$.

Many well-known distributions belong to the exponential family. For example, the Normal, Poisson, Binomial and Gamma distributions can all be written in the form (2.2).

2.2.1 Poisson distribution

The pf of the Poisson distribution with parameter θ is

$$f(y; \theta) = \frac{\theta^y e^{-\theta}}{y!} \quad (2.3)$$

where y can take on the values $0, 1, 2, \dots$. The pf can be written as

$$\begin{aligned} f(y; \theta) &= \exp[\log(\theta^y) - \theta - \log(y!)] \\ &= \exp[y \log(\theta) - \theta - \log(y!)] \end{aligned}$$

Comparing the above expression with (2.2) we see that

$$a(y) = y$$

$$\begin{aligned}
b(\theta) &= \log(\theta) \\
c(\theta) &= -\theta \\
d(y) &= -\log(y!)
\end{aligned}$$

The distribution is in canonical form, since $a(y) = y$, and its natural parameter is $b(\theta) = \log(\theta)$.

The Poisson distribution, denoted by $Y \sim \text{Poisson}(\theta)$, is used to model count data.

2.2.2 Normal distribution

The pdf of the Normal distribution is

$$f(y; \mu) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left[-\frac{1}{2\sigma^2}(y - \mu)^2 \right] \quad (2.4)$$

Here μ is the parameter of interest and σ^2 is regarded as a nuisance parameter. The pdf can be rewritten as

$$f(y; \mu) = \exp \left\{ y \cdot \frac{\mu}{\sigma^2} - \frac{\mu^2}{2\sigma^2} - \left[\frac{y^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma^2) \right] \right\} \quad (2.5)$$

Again comparing expression (2.5) with (2.2) we recognize that

$$\begin{aligned}
a(y) &= y \\
b(\mu) &= \frac{\mu}{\sigma^2} \\
c(\mu) &= -\frac{\mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \\
d(y) &= -\frac{y^2}{2\sigma^2}
\end{aligned}$$

The distribution is in canonical form, since $a(y) = y$, and its natural parameter is $b(\mu) = \mu/\sigma^2$.

The Normal distribution is used to model continuous data, particularly if the data arise from a symmetric distribution. The Normal distribution is widely used for at least three reasons. First, many natural phenomena are well described by the Normal distribution, such as height or blood pressure of people. Second, even if data are not Normally distributed (such as skewed data), the average of a random sample will be approximately Normally distributed (Central Limit Theorem). Third, much statistical theory and many statistical methods have been developed for the Normal distribution.

2.2.3 Binomial distribution

The pf of the Binomial distribution with parameter π (number of successes in n binary “trials”) is

$$f(y; \pi) = \binom{n}{y} \pi^y (1 - \pi)^{n-y} \quad (2.6)$$

where y can take the values $0, 1, 2, \dots, n$. We denote the distribution by $Y \sim \text{Bin}(\pi, n)$. Here π is the parameter of interest and n is usually assumed known. The pf can be rewritten as

$$\begin{aligned} f(y; \pi) &= \exp \left[y \log \pi - y \log(1 - \pi) + n \log(1 - \pi) + \log \binom{n}{y} \right] \\ &= \exp \left[y \log \left(\frac{\pi}{1 - \pi} \right) + n \log(1 - \pi) + \log \binom{n}{y} \right] \end{aligned}$$

Thus we have

$$\begin{aligned} a(y) &= y \\ b(\pi) &= \log \left(\frac{\pi}{1 - \pi} \right) \\ c(\pi) &= n \log(1 - \pi) \\ d(y) &= \log \binom{n}{y} \end{aligned}$$

The distribution is in canonical form and its natural parameter is $b(\pi) = \log[\pi/(1 - \pi)]$, which is the **log odds** or **logit** of π .

2.2.4 Gamma distribution

We use the parameterization of the Gamma distribution in terms of its mean μ and shape parameter ν (see McCullagh and Nelder, 1989 Chapter 8; SAS, 2004). Then, if the random variable Y has the Gamma distribution with parameter μ , which is the parameter of interest, and shape parameter ν (which is either known or a nuisance parameter), then its pdf is

$$f(y; \mu) = \frac{1}{\Gamma(\nu)y} \left(\frac{\nu y}{\mu} \right)^\nu \exp \left(-\frac{\nu y}{\mu} \right) \quad (2.7)$$

The pdf can be written as

$$f(y; \mu) = \exp \left[-\nu y \cdot \frac{1}{\mu} - \nu \log(\mu) + \nu [\log(y) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y) \right]$$

so that

$$\begin{aligned} a(y) &= -\nu y \\ b(\mu) &= \frac{1}{\mu} \\ c(\mu) &= -\nu \log(\mu) \\ d(y) &= \nu[\log(y) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y) \end{aligned}$$

The distribution is not in canonical form since $a(y) = -\nu y$, and its natural parameter is $b(\mu) = 1/\mu$.

2.3 Properties of Distributions in the Exponential Family

2.3.1 Mean and variance of $a(Y)$

For distributions in the exponential family convenient and useful expressions for the expected value and for the variance of $a(Y)$ can be derived. To find these expressions, we use the following results that apply for any pdf $f(y; \theta)$, provided that the order of differentiation and integration can be interchanged:

From the definition of a pdf, the area under the curve is 1, that is

$$\int f(y; \theta) dy = 1 \quad (2.8)$$

where integration is over the whole support of the distribution of Y (when the distribution is discrete, the integral is replaced by a sum). Differentiating both sides of (2.8) with respect to θ we obtain

$$\frac{d}{d\theta} \int f(y; \theta) dy = \frac{d}{d\theta} \cdot 1 = 0 \quad (2.9)$$

If the order of differentiation and integration in the first term in (2.9) can be reversed, which we assume, (2.9) becomes

$$\int \frac{df(y; \theta)}{d\theta} dy = 0 \quad (2.10)$$

Similarly, if (2.10) is differentiated twice with respect to θ and the order of differentiation and integration is reversed we obtain

$$\int \frac{d^2 f(y; \theta)}{d\theta^2} dy = 0 \quad (2.11)$$

These results can now be used for distributions in the exponential family. From expression (2.2) for the pdf of a distribution in the exponential family, namely

$$f(y; \theta) = \exp[a(y)b(\theta) + c(\theta) + d(y)]$$

we have

$$\frac{df(y; \theta)}{d\theta} = [a(y)b'(\theta) + c'(\theta)] f(y; \theta)$$

From (2.10)

$$\int [a(y)b'(\theta) + c'(\theta)] f(y; \theta) dy = 0$$

This can be simplified to

$$b'(\theta)E[a(Y)] + c'(\theta) = 0 \quad (2.12)$$

since $\int a(y)f(y; \theta) dy = E[a(Y)]$ by the definition of the expected value, and of course $\int f(y; \theta) dy = 1$. Rearranging (2.12) gives

$$E[a(Y)] = -\frac{c'(\theta)}{b'(\theta)} \quad (2.13)$$

Similarly, the variance of $a(Y)$ can be obtained:

$$\begin{aligned} \frac{d^2 f(y; \theta)}{d\theta^2} &= \frac{d}{d\theta} \{[a(y)b'(\theta) + c'(\theta)]f(y; \theta)\} \\ &= [a(y)b''(\theta) + c''(\theta)]f(y; \theta) + [a(y)b'(\theta) + c'(\theta)]^2 f(y; \theta) \end{aligned} \quad (2.14)$$

Using (2.13), the second term on the right hand side of expression (2.14) can be rewritten as

$$[b'(\theta)]^2 \{a(y) - E[a(Y)]\}^2 f(y; \theta)$$

Then, from (2.11),

$$\begin{aligned} \int \frac{d^2 f(y; \theta)}{d\theta^2} dy &= \int [a(y)b''(\theta) + c''(\theta)]f(y; \theta) dy + \int [b'(\theta)]^2 \{a(y) - E[a(Y)]\}^2 f(y; \theta) dy \\ &= b''(\theta)E[a(y)] + c''(\theta) + [b'(\theta)]^2 \text{Var}[a(y)] \\ &= 0 \end{aligned}$$

since $\int \{a(y) - E[a(Y)]\}^2 f(y; \theta) dy = \text{Var}[a(Y)]$ by definition. Rearranging and again substituting (2.13) gives

$$\text{Var}[a(Y)] = \frac{b''(\theta)c'(\theta) - c''(\theta)b'(\theta)}{[b'(\theta)]^3} \quad (2.15)$$

When the pdf is in canonical form, that is, $a(y) = y$, Expressions (2.13) and (2.15) directly provide the expected value and variance of random variables Y for distributions in the exponential family.

2.3.2 Mean and variance of the score statistic

In addition to expressions for the mean and variance of $a(Y)$, we need expressions for the expected value and variance of the derivative of the log-likelihood function. The likelihood function is the pdf (or pf), but written as a function of the parameter θ , given y . Generally we denote the likelihood by $L(\theta; y)$, and the logarithm of the likelihood by $l(\theta; y) = \log[L(\theta; y)]$. From (2.5) the log-likelihood for a distribution in the exponential family is

$$l(\theta; y) = a(y)b(\theta) + c(\theta) + d(y) \quad (2.16)$$

The derivative of $l(\theta; y)$ with respect to θ is

$$U(\theta; y) = \frac{dl(\theta; y)}{d\theta} = a(y)b'(\theta) + c'(\theta) \quad (2.17)$$

The function $U = U(\theta; y)$ is called the score statistic, and since it depends on y it can be regarded as a random variable. Its expected value is

$$E(U) = b'(\theta)E[a(Y)] + c'(\theta)$$

From (2.13), namely $E[a(Y)] = -c'(\theta)/b'(\theta)$, we have

$$E(U) = b'(\theta) \frac{-c'(\theta)}{b'(\theta)} + c'(\theta) = 0 \quad (2.18)$$

so that the expectation of the score statistic is zero for any distribution in the exponential family.

The variance of U is called the **information** and will be denoted by $\mathbf{I}$. From (2.17) we have

$$\mathbf{I} = \text{Var}(U) = \text{Var}[a(Y)b'(\theta) + c'(\theta)] = [b'(\theta)]^2 \text{Var}[a(Y)]$$

Substituting (2.15) gives

$$\mathbf{I} = \text{Var}(U) = \frac{b''(\theta)c'(\theta)}{b'(\theta)} - c''(\theta) \quad (2.19)$$

The score statistic U is used for inference about parameter values in a generalized linear model (see Chapter 4).

Another property of U which will be used later is

$$\text{Var}(U) = E(U^2) = -E(U') \quad (2.20)$$

so that the variance of the score statistic is minus 1 times the expected value of the derivative of the score. The first equality follows simply from the fact that $E(U) = 0$ (see Equation (2.18)). To show the second equality we differentiate U as in (2.17) with respect to θ :

$$U' = \frac{dU}{d\theta} = a(y)b''(\theta) + c''(\theta)$$

Therefore the expected value of U' is

$$\begin{aligned} E(U') &= b''(\theta)E[a(Y)] + c''(\theta) \\ &= b''(\theta)\frac{-c'(\theta)}{b'(\theta)} + c''(\theta) \\ &= -\text{Var}(U) = -\mathbf{I} \end{aligned}$$

2.4 Generalized Linear Models

Generalizing the conventional linear model with normally distributed error terms, Nelder and Wedderburn (1972) proposed a large class of statistical models using the concept of a **Generalized Linear Model**. The generalized linear model is defined in terms of a set of independent random variables $Y_1, \dots, Y_n$, each with a distribution from the exponential family and with the following properties:

1. The distribution of each Y_i has canonical form and depends on a single parameter θ_i (the θ_i do not all have to be the same for $i = 1, \dots, n$), thus

$$f(y_i; \theta_i) = \exp[y_i b_i(\theta_i) + c_i(\theta_i) + d_i(y_i)]$$

2. The distribution of all Y_i ($i = 1, \dots, n$) has the same form (for example, all Normal, all Binomial or all Poisson) so that the subscripts on b , c and d are not needed. Thus the joint pdf of $Y_1, \dots, Y_n$ is

$$\begin{aligned} f(y_1, \dots, y_n; \theta_1, \dots, \theta_n) &= \prod_{i=1}^n \exp[y_i b(\theta_i) + c(\theta_i) + d(y_i)] \\ &= \exp \left\{ \sum_{i=1}^n y_i b(\theta_i) + \sum_{i=1}^n c(\theta_i) + \sum_{i=1}^n d(y_i) \right\} \end{aligned}$$

The parameters θ_i are typically not of direct interest (there may be a different parameter for each observation). Instead, we specify a **model** for the $\theta_1, \dots, \theta_n$ in terms of a **smaller** set of parameters $\beta_1, \dots, \beta_p$, with $p < n$.

Suppose that $E(Y_i) = \mu_i$ where μ_i is some function of θ_i . Specifically, for a generalized **linear** model there is a transformation $g(\cdot)$ of μ_i such that

$$g(\mu_i) = x_i' \beta \quad (2.1)$$

Here g is a monotone differentiable function called the **link function** and x_i is a $p \times 1$ vector of explanatory variables (covariates and dummy variables for levels of factors) so that $x_i' = (x_{i1}, \dots, x_{ip})$ is the i -th row of the design matrix X .

Finally, β is the $p \times 1$ vector of model parameters, namely

$$\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

In summary, a generalized linear model has three components:

1. **Error structure / distribution:**

Independent response variables $Y_1, \dots, Y_n$ assumed to have the **same** distribution from the **exponential family**.

2. **Linear Predictor:**

A set of parameters β and explanatory variables

$$X = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}$$

such that

$$\eta = X\beta$$

3. Link Function:

A monotone link function g such that $g(\mu_i) = \eta_i = x'_i\beta$, $i = 1, \dots, n$, or in matrix notation

$$g(\mu) = \eta = X\beta$$

where $\mu = E(Y)$.

2.5 Examples

2.5.1 Normal linear model

The best known and most widely used special case of a generalized linear model is of course the conventional, or Normal, linear model

$$E(Y) = \mu = X\beta$$

where the $Y_1, \dots, Y_n$ are independent with $Y_i \sim N(\mu_i, \sigma^2)$. In matrix notation, the model is usually written in the form

$$y = X\beta + e$$

where $e = (e_1, \dots, e_n)'$ are independent and identically distributed error terms with $e_i \sim N(0, \sigma^2)$ for $i = 1, \dots, n$.

For the normal linear model, the three components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_1, \dots, Y_n$ with $Y_i \sim N(\mu_i, \sigma^2)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\mu = X\beta$.

3. Link Function:

The link function g is the identity function, since

$$g(\mu) \equiv \mu = X\beta$$

The SAS PROC GENMOD code to fit the ANCOVA model of Section 1.2.3 is given below. Here all specifications are written out explicitly in order to illustrate the three components of a generalized linear model:

```
PROC GENMOD DATA=sbp;
  TITLE 'Analysis of Covariance (ANCOVA)';
  CLASS centre treat;
  MODEL postsbp = presbp centre treat / DIST=normal LINK=identity;
RUN;
```

Thus, in SAS PROC GENMOD, the three components of a generalized linear model are specified as follows:

1. The error structure/distribution of the data is specified as an option in the **MODEL** statement; here: **DIST=normal**.
2. The linear predictor is specified in the **MODEL** statement as such; here:
MODEL postsbp = presbp centre treat.
3. The link function is specified as another option in the **MODEL** statement; here:
LINK=identity. (Note that the latter is not strictly required, since the identity link function is the SAS default for the Normal distribution.)

The output for the above SAS code is included with Appendix 2.1 (identical to Appendix 1.2).

2.5.2 Contingency tables / log-linear models

A cross-sectional study was carried out in October 2008 to investigate student attitudes at UFS to the resignation of the rector. A random sample of registered students was drawn from the total student population, and cross-classified according to gender, and whether they thought it was appropriate for the rector to resign or not. The (fictional!) data are summarized in the following two-way contingency table:

Table 2.1: Student Attitude to Resignation of Rector

A: Gender	B: Resignation appropriate		Total
	Yes	No	
Female	90	40	130
Male	80	40	120
Total	170	80	250

Let y_{ij} be the number of subjects in cell (ij) of the table. We can write

$$y_{ij} = \mu_{ij} + e_{ij}$$

where $Y_{ij} \sim \text{Poisson}(\mu_{ij})$.

If we assume that the variables A (Gender) and B (Resignation) are independent, we have

$$\mu_{ij} = n \cdot \pi_i \cdot \pi_j$$

where π_i is the probability of falling into the i -th category of variable A, and π_j is the probability of falling into the j -th category of variable B.

Now let

$$\begin{aligned}
 \eta_{ij} &= \log(\mu_{ij}) \\
 &= \log(n \cdot \pi_i \cdot \pi_j) \\
 &= \log(n) + \log(\pi_i) + \log(\pi_j) \\
 &= \theta + \lambda_i^A + \lambda_j^B
 \end{aligned}$$

The above is a **log-linear model** for the two-way table, Table 2.1. For the log-linear model, the three components of the generalized linear model are as follows:

1. **Error structure / Distribution:**

Independent response variables $Y_{ij} \sim \text{Poisson}(\mu_{ij})$.

2. **Linear Predictor:**

Parameters β and explanatory variables X such that $\eta_{ij} = \theta + \lambda_i^A + \lambda_j^B$, or, in matrix notation

$$\eta = X\beta$$

where X and $\beta = (\theta, \lambda_1^A, \lambda_2^A, \lambda_1^B, \lambda_2^B)'$ are the design matrix and parameter vector for a two-way layout.

3. **Link Function:**

The link function g is the logarithm, since $g(\mu_{ij}) = \log(\mu_{ij}) = \theta + \lambda_i^A + \lambda_j^B = x_i' \beta$.

In matrix notation:

$$g(\mu) = \log(\mu) = X\beta$$

The SAS PROC GENMOD code to fit the above so-called log-linear model (for more details, see Section 6.3 to 6.8 which are devoted to such models) is given below:

```
PROC GENMOD DATA=rector;
  TITLE 'Contingency Table';
  CLASS gender appropriate;
  MODEL count = gender appropriate / DIST=poisson LINK=log;
RUN;
```

The output for the above SAS code is included with Appendix 2.2. Again, the three components of a generalized linear model have been specified as follows:

1. The error structure/distribution of the data is specified as an option in the **MODEL** statement; here: **DIST=poisson**.
2. The linear predictor is specified in the **MODEL** statement as such; here:
MODEL count = gender appropriate.

3. The link function is specified as another option in the **MODEL** statement; here: **LINK=log**. (Note that again the latter is not strictly required, since the log link function is the SAS default for the Poisson distribution.)

Note that as with linear ANOVA models, the categorical model variables must be specified as such (being categorical) in the **CLASS** statement.

2.5.3 Probit analysis

Let Y_i be the number of successes in n_i independent trials where π_i is the probability in the i -th set of trials. Then we have $Y_i \sim \text{Bin}(\pi_i, n_i)$ and we can write

$$y_i = n_i \pi_i + e_i, \quad i = 1, \dots, N$$

Suppose that we want to model the proportion of experimental animals killed by various dose levels x of a toxic substance. The aim is to describe the probability of death, π , as a function of the dose.

A commonly used model for the relationship between π and x is the **Probit** model. Here it is assumed that each experimental animal has a **tolerance** y to the toxic substance: if the tolerance is smaller than the dose, $y < x$, the animal dies, otherwise, if $y \geq x$, the animal survives. Furthermore, it is assumed that the tolerance y has a distribution in the population of experimental animals. If we choose the Normal $N(\mu, \sigma^2)$ distribution as tolerance distribution, we obtain the Probit model. In this case, the probability π_i of an animal dying, when receiving the dose x_i , is given by

$$\begin{aligned} \pi_i &= \text{Prob}(Y < x_i), & \text{where } Y &\sim N(\mu, \sigma^2) \\ &= \int_{-\infty}^{x_i} (2\pi\sigma^2)^{-1/2} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2} \\ &= \Phi\left(\frac{x_i - \mu}{\sigma}\right) \end{aligned}$$

where Φ is the cdf of the Normal $N(0, 1)$ distribution.

If, in an experiment, n_i animals are exposed to doses x_i in $i = 1, \dots, N$ groups, and y_i animals die, we can write

$$\begin{aligned} E(Y_i) = \mu_i &= n_i \cdot \pi_i \\ &= n_i \cdot \Phi\left(\frac{x_i - \mu}{\sigma}\right) \end{aligned}$$

so that

$$\frac{\mu_i}{n_i} = \Phi\left(\frac{x_i - \mu}{\sigma}\right)$$

or, equivalently,

$$\begin{aligned}\Phi^{-1}\left(\frac{\mu_i}{n_i}\right) &= \frac{x_i - \mu}{\sigma} \\ &= -\frac{\mu}{\sigma} + x_i \cdot \frac{1}{\sigma} \\ &= \beta_1 + x_i \beta_2\end{aligned}$$

where $\beta_1 = -\mu/\sigma$ and $\beta_2 = 1/\sigma$. Therefore, for the probit model, the three components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_i \sim \text{Bin}(\pi_i, n_i)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = \beta_1 + x_i \beta_2$, or, in matrix notation

$$\eta = X\beta$$

where X and $\beta = (\beta_1, \beta_2)'$ are the design matrix and parameter vector for a simple linear regression.

3. Link Function:

The link function g is the Probit (inverse of the cdf of the standard Normal distribution), since $g(\mu_i) = \Phi^{-1}(\mu_i/n_i) = \beta_1 + x_i \beta_2$. In matrix notation:

$$g(\mu) = \Phi^{-1}(\mu/n) = X\beta$$

We illustrate the SAS code for a probit model using an example treated in more detail in Section 5.3.4. Table 5.1 in Section 5.3.4 shows the number of beetles dead after 5 hours of exposure to carbon disulphide gas at various concentrations (data from Bliss, 1935). We fit a probit model with $x_i = \log(\text{concentration}_i)$, using the following SAS code:

```

PROC GENMOD DATA=beetles;
    TITLE 'Probit Model for Beetle Mortality';
    MODEL y/n = logdose / DIST=binomial LINK=probit;
RUN;

```

Thus the three components of a generalized linear model have been specified as follows:

1. The error structure/distribution of the data is specified as an option in the **MODEL** statement; here: **DIST=binomial**.
2. The linear predictor is specified in the **MODEL** statement as such; here:
MODEL y/n = logdose.
3. The link function is specified as another option in the **MODEL** statement; here:
LINK=probit. Note that in the case of the probit model the specification of the link function is in fact required, since for the binomial distribution the logit link function is the SAS default.

2.6 Canonical Links

Each distribution in the exponential family has a special link function for which there exists a sufficient statistic equal in dimension to β in the linear predictor $\eta = X\beta$. These **canonical links** occur when

$$g(\mu) = \eta = b(\theta)$$

where $b(\theta)$ is the natural or canonical parameter of the distribution as defined in (2.2). The canonical link, therefore, links the mean of the distribution to its natural parameter. For the distributions discussed in Sections 2.2.1 to 2.2.4 the canonical links are as follows:

Normal	$g(\mu) = \mu$
Poisson	$g(\mu) = \log(\mu)$
Binomial	$g(\mu) = \log\left(\frac{\pi}{1-\pi}\right)$
Gamma	$g(\mu) = \mu^{-1}$

For the canonical links the sufficient statistic for β , in matrix notation, is $T(Y) = X'Y$, that is, the sufficient statistic is a linear function of the data. However, note that although the canonical links lead to desirable statistical properties of the model, the systematic effects in a model are not necessarily additive on the scale given by that link. It is convenient if they are, but convenience alone cannot replace quality of fit as a model selection criterion. There are instances (see, for example, the probit model in Section 2.5.3) in which non-canonical links are used. Nevertheless, canonical links are in fact often very sensible on scientific grounds (and not only on grounds of statistical convenience).

Note that in SAS PROC GENMOD the canonical link for a given distribution is the default link used when no link function is specified in the **MODEL** statement.

Chapter 3

Estimation

The present chapter is about obtaining point estimates of the parameters in generalized linear models, using methods based on maximum likelihood. Although explicit mathematical expressions for estimators can be found in some special cases, numerical methods for finding the estimates are usually required. Typically these methods are iterative and are based on the Newton-Raphson algorithm. Therefore, we begin by explaining the general principle of the Newton-Raphson algorithm; thereafter, we develop the theory of estimation in generalized linear models by applying the Newton-Raphson algorithm to those models.

3.1 Newton-Raphson Algorithm

In general, to find the maximum likelihood estimate for a parameter θ of a statistical model, we go through the following three steps:

1. Calculate the log of the likelihood
2. Take the derivative of the log-likelihood with respect to the parameter
3. Set the derivative equal to zero and solve for the parameter

Thus, if $l(\theta; y)$ is the log-likelihood, we must solve the equation

$$\frac{\partial l(\theta; y)}{\partial \theta} = 0$$

for θ .

The Newton-Raphson method is an iterative (numerical) method to find the solution $x^{(m)}$ to the equation

$$f(x) = 0 \quad (3.1)$$

Let us assume that $x^{(m-1)}$ is an approximate solution to Equation (3.1). Then, writing $x^{(m)} = x^{(m-1)} + h$, and using a first order Taylor expansion of $f(x)$ about $x^{(m-1)}$, we can write

$$\begin{aligned} 0 = f(x^{(m)}) &= f(x^{(m-1)} + h) \\ &\approx f(x^{(m-1)}) + hf'(x^{(m-1)}) \\ &= f(x^{(m-1)}) + (x^{(m)} - x^{(m-1)})f'(x^{(m-1)}) \end{aligned}$$

Therefore

$$x^{(m)} \approx x^{(m-1)} - \frac{f(x^{(m-1)})}{f'(x^{(m-1)})}$$

The above is the Newton-Raphson “update” formula for solving $f(x) = 0$. Starting with an initial guess $x^{(1)}$, successive “updates” $x^{(m)}$ to the current approximation $x^{(m-1)}$ are obtained using the formula

$$x^{(m)} = x^{(m-1)} - \frac{f(x^{(m-1)})}{f'(x^{(m-1)})} \quad (3.2)$$

until the iterative process converges.

As pointed out above, in the case of maximum likelihood estimation of a parameter θ , we must solve the equation

$$\frac{\partial l(\theta; y)}{\partial \theta} = U(\theta) = 0$$

where U is the score function. The update equation equivalent to (3.2) is therefore

$$\theta^{(m)} = \theta^{(m-1)} - \frac{U(\theta^{(m-1)})}{U'(\theta^{(m-1)})} \quad (3.3)$$

For maximum likelihood estimation, it is common to approximate U' in (3.3) by its expectation, $E(U')$. For distributions in the exponential family we have shown in Section 2.3.2 that $E(U') = -\mathbf{I}$, where $\mathbf{I} = \text{Var}(U)$ is the information. Therefore, an alternative to update equation (3.3) is

$$\theta^{(m)} = \theta^{(m-1)} + \frac{U(\theta^{(m-1)})}{\mathbf{I}(\theta^{(m-1)})} \quad (3.4)$$

Equation (3.4) is called the **method of scoring**.

In the next section, the Newton-Raphson and scoring methods for general maximum likelihood estimation presented above are developed for generalized linear models.

3.2 Estimation in Generalized Linear Models

We consider independent random variables $Y_1, \dots, Y_n$ satisfying the properties of a generalized linear model (see Section 2.4). We want to estimate the parameters β which are linked to the observations Y_i through the relationships $E(Y_i) = \mu_i$ and $g(\mu_i) = x_i' \beta$.

For each Y_i the log-likelihood is

$$l_i = y_i b(\theta_i) + c(\theta_i) + d(y_i) \quad (3.5)$$

where the functions b , c and d are defined in (2.2). Furthermore (using the results of Section 2.3.1 and noting that $a(y_i) = y_i$

$$E(Y_i) = \mu_i = -\frac{c'(\theta_i)}{b'(\theta_i)} \quad (3.6)$$

$$\text{Var}(Y_i) = \frac{b''(\theta) c'(\theta) - c''(\theta) b'(\theta)}{[b'(\theta)]^3} \quad (3.7)$$

$$g(\mu_i) = x_i' \beta = \eta_i \quad (3.8)$$

where x_i is a vector with elements x_{ij} , $j = 1, \dots, p$.

The likelihood function for all the Y_i is

$$l = \sum_{i=1}^n l_i = \sum_{i=1}^n y_i b(\theta_i) + \sum_{i=1}^n c(\theta_i) + \sum_{i=1}^n d(y_i)$$

To obtain the maximum likelihood estimator for β_j we need the derivative of l with respect to β_j , namely, using the chain rule for differentiation

$$\frac{\partial l}{\partial \beta_j} = U_j = \sum_{i=1}^n \frac{\partial l_i}{\partial \beta_j} = \sum_{i=1}^n \left[\frac{\partial l_i}{\partial \theta_i} \cdot \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \beta_j} \right] \quad (3.9)$$

We now consider each term on the right hand side of (3.9) separately. Firstly

$$\begin{aligned} \frac{\partial l_i}{\partial \theta_j} &= y_i b'(\theta_i) + c'(\theta_i) && \text{by differentiating (3.5)} \\ &= y_i b'(\theta_i) - \mu_i b'(\theta_i) && \text{by solving (3.6) for } c'(\theta_i) \\ &= b'(\theta_i)(y_i - \mu_i) && \end{aligned} \quad (3.10)$$

Secondly

$$\frac{\partial \theta_i}{\partial \mu_i} = \left(\frac{\partial \mu_i}{\partial \theta_i} \right)^{-1}$$

Differentiating (3.6) yields

$$\begin{aligned} \frac{\partial \mu_i}{\partial \theta_i} &= -\frac{c''(\theta_i)}{b'(\theta_i)} + \frac{c'(\theta_i)b''(\theta_i)}{[b'(\theta_i)]^2} \\ &= b'(\theta_i)\text{Var}(Y_i) \quad \text{from (3.7)} \end{aligned} \quad (3.11)$$

Finally, from (3.8)

$$\begin{aligned} \frac{\partial \mu_i}{\partial \beta_j} &= \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j} \\ &= \frac{\partial \mu_i}{\partial \eta_i} \cdot x_{ij} \end{aligned} \quad (3.12)$$

Using (3.10)–(3.12) in (3.9) the score is given by

$$U_j = \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] \quad (3.13)$$

The covariance matrix of the U_j 's has terms

$$\mathbf{I}_{jk} = \text{E}[U_j U_k]$$

which form the **information matrix** $\mathbf{I}$. From (3.13) we have

$$\begin{aligned} \mathbf{I}_{jk} &= \text{E} \left\{ \sum_{i=1}^n \left[\frac{(Y_i - \mu_i)}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] \sum_{l=1}^n \left[\frac{(Y_l - \mu_l)}{\text{Var}(Y_l)} x_{lk} \left(\frac{\partial \mu_l}{\partial \eta_l} \right) \right] \right\} \\ &= \sum_{i=1}^n \frac{\text{E}[(Y_i - \mu_i)^2] x_{ij} x_{ik}}{[\text{Var}(Y_i)]^2} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \end{aligned} \quad (3.14)$$

because $\text{E}[(Y_i - \mu_i)(Y_l - \mu_l)] = 0$ for $i \neq l$ since the Y_i are independent. Noting that $\text{E}[(Y_i - \mu_i)^2] = \text{Var}(Y_i)$, Equation (3.14) can be simplified to

$$\mathbf{I}_{jk} = \sum_{i=1}^n \frac{x_{ij} x_{ik}}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \quad (3.15)$$

The updating equation (3.4) for the method of scoring can therefore be written as

$$\beta^{(m)} = \beta^{(m-1)} + [\mathbf{I}^{(m-1)}]^{-1} U^{(m-1)} \quad (3.16)$$

where $\beta^{(m)}$ is the vector of estimates of the parameters $\beta = [\beta_1, \dots, \beta_p]'$ at the m -th iteration. In (3.16), $[\mathbf{I}^{(m-1)}]^{-1}$ is the inverse of the information matrix with elements

$\mathbf{I}_{jk}$ given by (3.15) and $U^{(m-1)}$ is the vector of scores with elements given by (3.13). Multiplying both sides of Equation (3.16) by $\mathbf{I}^{(m-1)}$ we obtain

$$\mathbf{I}^{(m-1)}\beta^{(m)} = \mathbf{I}^{(m-1)}\beta^{(m-1)} + U^{(m-1)} \quad (3.17)$$

From (3.15) $\mathbf{I}$ can be written as

$$\mathbf{I} = X'WX \quad (3.18)$$

where W is the $n \times n$ diagonal matrix with elements

$$w_{ii} = \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \quad (3.19)$$

The right hand side of (3.17) is the vector with elements

$$\sum_{k=1}^p \sum_{i=1}^n \frac{x_{ij}x_{ik}}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 \beta_k^{(m-1)} + \sum_{i=1}^n \frac{(y_i - \mu_i)x_{ij}}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)$$

evaluated at $\beta^{(m-1)}$, which follows from Equations (3.15) and (3.13). Thus the right hand side of Equation (3.17) can be written as

$$X'Wz$$

where the **modified dependent variable** z has elements

$$z_i = \sum_{k=1}^p x_{ik}\beta_k^{(m-1)} + (y_i - \mu_i) \left(\frac{\partial \eta_i}{\partial \mu_i} \right) \quad (3.20)$$

with μ_i and $\partial \eta_i / \partial \mu_i$ evaluated at $\beta^{(m-1)}$. Therefore, z can in turn be written as

$$\begin{aligned} z &= X\beta^{(m-1)} + (y - \mu) \left(\frac{\partial \eta}{\partial \mu} \right) \\ &= \eta^{(m-1)} + (y - \mu) \left(\frac{\partial \eta}{\partial \mu} \right) \end{aligned}$$

Hence the update equation (3.17) can be written as

$$X'WX\beta^{(m)} = X'Wz \quad (3.21)$$

or

$$\beta^{(m)} = (X'WX)^{-1}X'Wz$$

Equation (3.21) is of the same form as the **normal equations** for a linear model obtained by weighted least squares, except that the normal equations have to be solved

iteratively because, in general, the weights W and the modified dependent variable z depend on β . Thus, for generalized linear models, maximum likelihood estimators can be obtained through an **iteratively reweighted least squares procedure**.

Note:

The import of Equation (3.21), in conjunction with the expressions for the weight matrix W in (3.19) and for the modified dependent variable z in (3.20), is the following: In order to obtain the ML estimates for β we do not need to derive explicitly the score U and the information matrix $\mathbf{I}$ using an expression for the log-likelihood and the chain rule (3.9). Instead, can obtain the ML estimates for β directly by solving the normal equations (3.21), for which we need to determine only the following three elements:

1. The design matrix X , which is unproblematic.
2. The diagonal matrix of weights W , given by (3.19). In order to evaluate (3.19) we need to determine
 - (a) $\text{Var}(Y_i)$ for the distribution in question (we can get this from (2.15))
 - (b) The derivative $\partial\mu_i/\partial\eta_i$, which depends on the specified link function.
3. The modified dependent variable z , given by (3.20). In order to evaluate (3.20) we need to determine $\partial\eta_i/\partial\mu_i$, which is simply the inverse of $\partial\mu_i/\partial\eta_i$ determined under 2(a) above.

Statistical packages that include procedures for fitting generalized linear models have algorithms based on the normal equations (3.21). They begin by using some initial approximation $\beta^{(0)}$ to evaluate z and W , then (3.21) is solved to provide an updated estimate $\beta^{(1)}$. Then in turn $\beta^{(1)}$ is used to obtain better approximations for z and W , and so on, until convergence is achieved. When the difference between successive approximations $\beta^{(m-1)}$ and $\beta^{(m)}$ is sufficiently small, the iteration is stopped and $\beta^{(m)}$ is taken as the maximum likelihood estimate for β .

3.3 Examples: Maximum Likelihood Estimation

3.3.1 Poisson regression with log-link

We have a sample $y_1, \dots, y_n$ from the Poisson distribution, and a corresponding covariate $x_1, \dots, x_n$. The components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_1, \dots, Y_n \sim \text{Poisson}(\mu_i)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = \beta_0 + \beta_1 x_i$, or, in matrix notation

$$\eta = X\beta$$

where X and $\beta = (\beta_0, \beta_1)'$ are the design matrix and parameter vector for a simple linear regression as in Section 1.4.1 (Equation (1.12)).

3. Link Function:

The link function g is the logarithm, namely $g(\mu_i) = \log(\mu_i) = \eta_i = \beta_0 + \beta_1 x_i$.

In matrix notation:

$$g(\mu) = \log(\mu) = \eta = X\beta$$

The log-likelihood for a sample from the Poisson distribution is given by

$$l(\mu; y) = \sum_{i=1}^n [y_i \log(\mu_i) - \mu_i + \log(y_i!)]$$

Generally, in order to find the MLE for β , we need to find $U_j = \partial l / \partial \beta_j$, $j = 0, \dots, p$, set the derivative equal to zero, and solve the resulting set of equations for β . Using the chain rule for differentiation

$$\frac{\partial l}{\partial \beta_j} = U_j = \sum_{i=1}^n \frac{\partial l_i}{\partial \beta_j} = \sum_{i=1}^n \left[\frac{\partial l_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j} \right]$$

But

$$\begin{aligned}
\frac{\partial l_i}{\partial \mu_i} &= \frac{y_i}{\mu_i} - 1 \\
\frac{\partial \mu_i}{\partial \eta_i} &= e^\eta && \text{since } \log(\mu) = \eta \iff \mu = e^\eta \\
\frac{\partial \eta_i}{\partial \beta_0} &= 1 && \text{since } \eta_i = \beta_0 + \beta_1 x_i \\
\frac{\partial \eta_i}{\partial \beta_1} &= x_i
\end{aligned}$$

Thus we can calculate the score from “first principles” as

$$\begin{aligned}
U_j &= \sum_{i=1}^n \left[\frac{\partial l_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j} \right] \\
&= \sum_{i=1}^n \left[\left(\frac{y_i}{\mu_i} - 1 \right) \cdot e^\eta \cdot x_{ij} \right] \\
&= \sum_{i=1}^n \left[\frac{y_i - \mu_i}{\mu_i} \cdot e^\eta \cdot x_{ij} \right] \\
&= \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] && \text{since } \text{Var}(Y_i) = \mu_i
\end{aligned}$$

which is indeed in agreement with the general result (3.13) for the score from a generalized linear model.

However, as pointed out at the end of Section 3.2, in order to obtain the ML estimates for β we do not need to derive the score U and the information matrix $\mathbf{I}$ from first principles. We obtain the ML estimates by solving the normal equations (3.21)

$$X'W^{(m-1)}X\beta^{(m)} = X'W^{(m-1)}z^{(m-1)}$$

where, for the present example (Poisson regression), X is the design matrix for a simple linear regression. Thus we only need to determine W , which is the diagonal matrix of weights, and the modified dependent variable z .

We note that for the Poisson(μ_i) distribution, we have

$$\text{Var}(Y_i) = \mu_i$$

and $\partial \mu_i / \partial \eta_i$ has been already derived above for the log link. Now we can find W and z from expressions (3.19) and (3.20), respectively.

First, the matrix W has elements

$$\begin{aligned}
 w_{ii}^{(m-1)} &= \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 && \text{from (3.19)} \\
 &= \frac{1}{\mu_i} (e^{\eta_i})^2 \\
 &= e^{\eta_i} && \text{since } \mu_i = e^{\eta_i} \\
 &= e^{\beta_0^{(m-1)} + x_i \beta_1^{(m-1)}}
 \end{aligned}$$

Finally, the modified dependent variable $z^{(m-1)}$ has elements

$$\begin{aligned}
 z_i^{(m-1)} &= \eta_i + (y_i - \mu_i) \left(\frac{\partial \eta_i}{\partial \mu_i} \right) && \text{from (3.20)} \\
 &= \eta_i + \frac{y_i - \mu_i}{\mu_i} && \text{since } \partial \eta_i / \partial \mu_i = 1/\mu_i \\
 &= \eta_i + \frac{y_i - e^{\eta_i}}{e^{\eta_i}} && \text{since } \mu_i = e^{\eta_i} \\
 &= \beta_0^{(m-1)} + x_i \beta_1^{(m-1)} + \frac{y_i - e^{\beta_0^{(m-1)} + x_i \beta_1^{(m-1)}}}{e^{\beta_0^{(m-1)} + x_i \beta_1^{(m-1)}}}
 \end{aligned}$$

3.3.2 Logistic regression with logit-link

We have a sample $y_1, \dots, y_n$ from the Binomial $\text{Bin}(\pi, 1)$ distribution, and a corresponding covariate $x_1, \dots, x_n$. The components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_1, \dots, Y_n \sim \text{Bin}(\pi_i, 1)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = \beta_0 + \beta_1 x_i$, or, in matrix notation

$$\eta = X\beta$$

where X and $\beta = (\beta_0, \beta_1)'$ are the design matrix and parameter vector for a simple linear regression.

3. Link Function:

The link function g is the logit, namely

$$g(\pi_i) = g(\mu_i) = \log \left(\frac{\mu_i}{1 - \mu_i} \right) = \eta_i = \beta_0 + \beta_1 x_i$$

In order to do logistic regression we solve the normal equations (3.21)

$$X'W^{(m-1)}X\beta^{(m)} = X'W^{(m-1)}z^{(m-1)}$$

where X is the design matrix for a simple linear regression. Thus we need to determine W , which is the diagonal matrix of weights, and the modified dependent variable z .

We note that for the $\text{Bin}(\pi_i, 1)$ distribution, we have

$$\text{Var}(Y_i) = \mu_i(1 - \mu_i)$$

and for the logit link,

$$\frac{\partial \mu_i}{\partial \eta_i} = \frac{e^{\eta_i}}{(1 + e^{\eta_i})^2}$$

since

$$\log \left(\frac{\mu}{1 - \mu} \right) = \eta \iff \mu = \frac{e^\eta}{1 + e^\eta}$$

Now we can find W and z from expressions (3.19) and (3.20), respectively. First, the matrix W has elements

$$\begin{aligned} w_{ii}^{(m-1)} &= \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 && \text{from (3.19)} \\ &= \frac{1}{\mu_i(1 - \mu_i)} \left[\frac{e^{\eta_i}}{(1 + e^{\eta_i})^2} \right]^2 && \text{since } \text{Var}(Y_i) = \mu_i(1 - \mu_i) \\ &= \frac{(1 + e^{\eta_i})^2}{e^{\eta_i}} \left[\frac{e^{\eta_i}}{(1 + e^{\eta_i})^2} \right]^2 && \text{since } \mu_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}} \\ &= \frac{e^{\eta_i}}{(1 + e^{\eta_i})^2} \\ &= \frac{e^{\beta_0^{(m-1)} + x_i \beta_1^{(m-1)}}}{\left(1 + e^{\beta_0^{(m-1)} + x_i \beta_1^{(m-1)}} \right)^2} \end{aligned}$$

Finally, the modified dependent variable $z^{(m-1)}$ has elements

$$\begin{aligned} z_i^{(m-1)} &= \eta_i + (y_i - \mu_i) \left(\frac{\partial \eta_i}{\partial \mu_i} \right) \quad \text{from (3.20)} \\ &= \eta_i + \left(y_i - \frac{e^{\eta_i}}{1 + e^{\eta_i}} \right) \frac{(1 + e^{\eta_i})^2}{e_i^\eta} \end{aligned}$$

where $z_i^{(m-1)}$ is evaluated at $\eta_i^{(m-1)} = \beta_0^{(m-1)} + x_i \beta_1^{(m-1)}$.

3.3.3 Normal linear model with identity link

We have sample $y_1, \dots, y_n$ from the Normal $N(\mu_i, \sigma^2)$ distribution, and want to fit a general linear model of the type (1.1). The components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_1, \dots, Y_n \sim N(\mu_i, \sigma^2)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$, or, in matrix notation

$$\eta = X\beta$$

where X and $\beta = (\beta_0, \beta_1, \dots, \beta_p)'$ are the design matrix and parameter vector for the general linear model.

3. Link Function:

The link function g is the identity function, namely

$$g(\mu_i) = \mu_i = \eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$

In order to fit the general linear model we solve the normal equations (3.21)

$$X'W^{(m-1)}X\beta^{(m)} = X'W^{(m-1)}z^{(m-1)}$$

where X is the design matrix for the general linear model. Thus we need to determine W , which is the diagonal matrix of weights, and the modified dependent variable z .

We note that for the $N(\mu_i, \sigma^2)$ distribution, we have

$$\text{Var}(Y_i) = \sigma^2$$

and for the identity link,

$$\frac{\partial \mu_i}{\partial \eta_i} = 1$$

Now we can find W and z from expressions (3.19) and (3.20), respectively. First, the matrix W has elements

$$\begin{aligned} w_{ii}^{(m-1)} &= \frac{1}{\text{Var}(Y_i)} \left(\frac{\partial \mu_i}{\partial \eta_i} \right)^2 && \text{from (3.19)} \\ &= \frac{1}{\sigma^2} \end{aligned}$$

Finally, the modified dependent variable $z^{(m-1)}$ has elements

$$\begin{aligned} z_i^{(m-1)} &= \eta_i + (y_i - \mu_i) \left(\frac{\partial \eta_i}{\partial \mu_i} \right) && \text{from (3.20)} \\ &= \eta_i + (y_i - \eta_i) \cdot 1 \\ &= y_i \end{aligned}$$

The normal equations are therefore given by

$$\begin{aligned} X'W^{(m-1)}X\beta^{(m)} &= X'W^{(m-1)}z^{(m-1)} \\ \iff X' \left(\frac{1}{\sigma^2} I_n \right) X\beta^{(m)} &= X' \left(\frac{1}{\sigma^2} I_n \right) y \\ \iff X'X\beta &= X'y \end{aligned} \tag{3.22}$$

The weight matrix W does not depend on β , nor does $z = y$, so that the superscript for β can be dropped in (3.22). Thus the well-known normal equations for the general linear model, (3.22), are obtained from the theory of generalized linear models when we assume a Normal distribution for the data and use the identity link. Of course, (3.22) can be solved without iteration and the ML estimates $\hat{\beta}$ for β are given by

$$\hat{\beta} = (X'X)^{-1}X'y \tag{3.23}$$

when X is of full rank. (When X is not of full rank, the inverse $(X'X)^{-1}$ must be replaced by a generalized inverse in (3.23).)

Chapter 4

Inference

4.1 Introduction

The two main tools of statistical inference are **hypothesis tests** and **confidence intervals** (or interval estimates). This chapter covers their derivation for generalized linear models.

Confidence intervals for model parameters are increasingly regarded as more useful than hypothesis tests about parameters, because the width of the confidence interval provides a measure of the precision with which inferences can be made. Specifically, a hypothesis test for a parameter, or for a contrast of parameters, only tells us whether the parameter “significantly” differs from a certain null value (often zero). A confidence interval for the same parameter, or for a contrast of parameters, is usually more informative: a confidence interval gives us a range of “plausible” values for the parameter (or parameter contrast) in question. In fact, in medical statistics at least, scientific journals and regulatory authorities that review the statistical analyses of clinical trials require that, whenever possible, results of statistical analyses **must** be expressed in terms of confidence intervals.

In a modeling framework, hypothesis tests are performed by comparing how well two related models fit the data. For generalized linear models, the two models should have the same probability distribution and the same link function, but the linear component of the one model has more parameters than the other. The simpler model, corresponding to the null hypothesis H_0 , must be a special case of the more general (or

larger) model. The simpler model is said to be **nested** in the larger model. To compare the two models, we use summary statistics to describe how well the two models fit the data. The **goodness of fit** statistics are often based on one of the following

- the maximum value of the likelihood function
- the maximum value of the log-likelihood function
- the minimum value of a sum of squares criterion
- a composite statistic based on residuals

The process of goodness of fit testing can be summarized as follows:

1. Specify a statistical model M_0 corresponding to H_0 . Specify a model M_1 with M_0 nested in M_1 .
2. Fit both M_0 and M_1 and calculate the goodness of fit statistics G_0 and G_1 .
3. Calculate the improvement in fit, usually $G_1 - G_0$, but G_1/G_2 is another possibility.
4. Use the sampling distribution of $G_1 - G_0$ under H_0 to test the null hypothesis H_0 .

For both forms of inference, interval estimation and hypothesis testing, sampling distributions are required. To test a hypothesis, the sampling distribution of the goodness of fit statistic is required. To calculate a confidence interval, the sampling distribution of the estimator is required.

If the data are Normally distributed, the sampling distributions can often be determined exactly. For other distributions, we need to rely on large sample asymptotic results based on the central limit theorem. For independent observations from generalized linear models, the necessary regularity conditions for the validity of the central limit theorem are satisfied.

The basic idea is that, under appropriate conditions, if S is a statistic of interest, then approximately

$$\frac{S - E(S)}{\sqrt{\text{Var}(S)}} \sim N(0, 1)$$

where $E(S)$ and $\text{Var}(S)$ are respectively the expectation and variance of S . Equivalently to the above we have

$$\frac{[S - E(S)]^2}{\text{Var}(S)} \sim \chi_1^2$$

If there is a vector of statistics, namely $S = (S_1, \dots, S_p)'$ with asymptotic expectation $E(S)$ and asymptotic $p \times p$ covariance matrix $\text{Var}(S) = V$, then approximately

$$[S - E(S)]'V^{-1}[S - E(S)] \sim \chi_p^2 \quad (4.1)$$

provided that V is non-singular and therefore V^{-1} exists.

4.2 Sampling Distribution of the Score Statistic

Suppose $Y_1, \dots, Y_n$ are independent random variables in a generalized linear model with parameters β , where, as usual, $E(Y_i) = \mu_i$ and $g(\mu_i) = x_i'\beta = \eta_i$. From Equation (3.13) the score statistics are

$$U_j = \sum_{i=1}^n \left[\frac{(y_i - \mu_i)}{\text{Var}(Y_i)} x_{ij} \left(\frac{\partial \mu_i}{\partial \eta_i} \right) \right] \quad \text{for } j = 1, \dots, p$$

Since $E(Y_i) = \mu_i$ for all i , we have

$$E(U_j) = 0 \quad \text{for } j = 1, \dots, p \quad (4.2)$$

which is consistent with the general result (2.18). The covariance matrix of U is the information matrix $\mathbf{I}$ with elements

$$\mathbf{I}_{jk} = E(U_j U_k)$$

given by Equation (3.15).

If there is only one parameter, the score statistic has the asymptotic sampling distribution

$$\frac{U}{\sqrt{\mathbf{I}}} \sim N(0, 1), \quad \text{or equivalently} \quad \frac{U^2}{\mathbf{I}} \sim \chi_1^2$$

If there is a vector of parameters $\beta = (\beta_1, \dots, \beta_p)'$ then the score vector $U = (U_1, \dots, U_p)'$ has the multivariate Normal distribution $U \sim N(0, \mathbf{I})$, at least asymptotically, and therefore

$$U' \mathbf{I}^{-1} U \sim \chi_p^2 \quad (4.3)$$

for large samples.

4.3 Examples: Score Statistic

4.3.1 Score statistic for the Normal distribution

Let $Y_1, \dots, Y_n$ be independent and identically distributed random variables with $Y_i \sim N(\mu, \sigma^2)$ where σ^2 is a known constant. The log-likelihood function is

$$l = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 - n \log(\sigma\sqrt{2\pi})$$

The score statistic is

$$U = \frac{dl}{d\mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \mu) = \frac{n}{\sigma^2} (\bar{Y} - \mu)$$

Therefore the ML estimator for μ , obtained by solving the equation $U = 0$, is $\hat{\mu} = \bar{Y}$.

The expected value of the score statistic is

$$E(U) = \frac{1}{\sigma^2} \sum_{i=1}^n [E(Y_i) - \mu]$$

Since $E(Y_i) = \mu$ for all i , it follows that $E(U) = 0$ as expected. The variance of U is

$$\mathbf{I} = \text{Var}(U) = \frac{1}{\sigma^4} \sum_{i=1}^n \text{Var}(Y_i) = \frac{n}{\sigma^2}$$

since $\text{Var}(Y_i) = \sigma^2$. Therefore

$$\begin{aligned} \frac{U}{\sqrt{\mathbf{I}}} &= \frac{(\bar{Y} - \mu)}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \text{or equivalently} \\ \frac{U^2}{\mathbf{I}} &= \frac{(\bar{Y} - \mu)^2}{\sigma^2/n} \sim \chi_1^2 \end{aligned}$$

These results do not only hold asymptotically, but are in fact exact, since $\bar{Y}$ has Normal distribution $\bar{Y} \sim N(\mu, \sigma^2/n)$ exactly.

The sampling distribution of U can be used to make inferences about μ . For example, an (exact) 95% confidence interval for μ is given by $\bar{y} \pm 1.96\sigma/\sqrt{n}$, when σ is known.

4.3.2 Score statistic for the Binomial distribution

If $Y \sim \text{Bin}(\pi, n)$ the log-likelihood function is

$$l(\pi; y) = y \log(\pi) + (n - y) \log(1 - \pi) + \log \binom{n}{y}$$

Therefore the score statistic is

$$U = \frac{dl}{d\pi} = \frac{Y}{\pi} - \frac{n - Y}{1 - \pi} = \frac{Y - n\pi}{\pi(1 - \pi)}$$

But $E(Y) = n\pi$ and so $E(U) = 0$, as expected. Furthermore, $\text{Var}(Y) = n\pi(1 - \pi)$, so

$$\mathbf{I} = \text{Var}(U) = \frac{1}{\pi^2(1 - \pi)^2} \cdot \text{Var}(Y) = \frac{n}{\pi(1 - \pi)}$$

Therefore, approximately,

$$\frac{U}{\sqrt{\mathbf{I}}} = \frac{Y - n\pi}{\sqrt{n\pi(1 - \pi)}} \sim N(0, 1)$$

This is the Normal approximation to the Binomial distribution (without continuity correction) which is used to find approximate confidence intervals for and test hypotheses about π .

4.4 Taylor Series Approximations

To obtain asymptotic distributions for various statistics other than the score statistics, it is often useful to use Taylor series approximations. The Taylor series approximation for a function $f(x)$ of a single variable x about a value t is

$$f(x) = f(t) + (x - t)f'(t) + \frac{1}{2}(x - t)^2f''(t) + \dots$$

For a log-likelihood function of a single parameter β the first three terms of a Taylor series approximation near an estimate b for β are

$$\begin{aligned} l(\beta) &= l(b) + (\beta - b)l'(b) + \frac{1}{2}(\beta - b)^2l''(b) \\ &= l(b) + (\beta - b)U(b) + \frac{1}{2}(\beta - b)^2U'(b) \end{aligned}$$

where $U(b) = dl/d\beta$ is the score function evaluated at the estimate b for β . If $U' = d^2l/d\beta^2$ is approximated by its expectation $E(U') = -\mathbf{I}$, the approximation becomes

$$l(\beta) = l(b) + (\beta - b)U(b) - \frac{1}{2}(\beta - b)^2\mathbf{I}(b)$$

where $\mathbf{I}(b)$ is the information evaluated at $\beta = b$. The corresponding approximation for the log-likelihood function of a vector of parameters β is

$$l(\beta) = l(b) + (\beta - b)'U(b) - \frac{1}{2}(\beta - b)'\mathbf{I}(b)(\beta - b) \quad (4.4)$$

where U is now the vector of scores and $\mathbf{I}$ is the information matrix.

Similarly, for the score function of a single parameter β the first two terms of a Taylor series approximation near an estimate b give

$$U(\beta) = U(b) + (\beta - b)U'(b)$$

If again U' is approximated by its expectation $E(U') = -\mathbf{I}$, we obtain

$$U(\beta) = U(b) - (\beta - b)\mathbf{I}(b)$$

The corresponding expression for a vector of parameters β is

$$U(\beta) = U(b) - \mathbf{I}(b)(\beta - b) \quad (4.5)$$

4.5 Sampling Distribution of ML Estimators

Equation (4.5) can be used to obtain the sampling distribution of the ML estimator $b = \hat{\beta}$ for β . By definition, b is the estimator that maximizes $l(\beta)$ and so $U(b) = 0$. Therefore, if b is the ML estimator, (4.5) becomes

$$U(\beta) = -\mathbf{I}(b)(\beta - b)$$

or equivalently

$$(b - \beta) = \mathbf{I}^{-1}U$$

if $\mathbf{I}$ is non-singular. If $\mathbf{I}$ is regarded as a constant, then $E(b - \beta) = 0$ because $E(U) = 0$ by Equations (4.2) and (2.18). Therefore $E(b) = \beta$, at least asymptotically, so b is a consistent estimator for β . The covariance matrix of b is

$$\begin{aligned} E[(b - \beta)'(b - \beta)] &= E[\mathbf{I}^{-1}UU^T\mathbf{I}^{-1}] \\ &= \mathbf{I}^{-1}E(UU^T)\mathbf{I}^{-1} \\ &= \mathbf{I}^{-1} \end{aligned} \quad (4.6)$$

because $E(UU^T) = \mathbf{I}$ (here U^T denotes the transpose of U in order to distinguish it from U' , which denotes the derivative).

By (4.1), therefore, the asymptotic sampling distribution of b is

$$(b - \beta)' \mathbf{I}(b)(b - \beta) \sim \chi_p^2 \quad (4.7)$$

Using expression (3.18) for the information matrix for a generalized linear model, and writing $\hat{\beta}$ for b , we have

$$(\hat{\beta} - \beta)' X' W X (\hat{\beta} - \beta) \sim \chi_p^2 \quad (4.8)$$

Statistics (4.7) and (4.8) are the **Wald statistic**.

A frequently used form of (4.7) is

$$b \sim N(\beta, \mathbf{I}^{-1}) \quad (4.9)$$

or, using again expression (3.18) for the information matrix for a generalized linear model,

$$\hat{\beta} \sim N\left(\beta, (X' W X)^{-1}\right) \quad (4.10)$$

If the response variables in the generalized linear model are Normally distributed (and if the identity link is used), then (4.7) – (4.10) are exact results.

4.5.1 Wald tests

The Wald statistic (4.8) and the result (4.10) can be used to test hypotheses about any linear contrast $L'\beta$ of β , namely

$$H_0 : L'\beta = 0 \quad \text{against} \quad H_1 : L'\beta \neq 0$$

where L is a $p \times c$ matrix of c linearly independent contrasts. Using (4.10), and since $L'\beta = 0$ under H_0 , the statistic

$$\hat{\beta}' L \left[L' (X' W X)^{-1} L \right]^{-1} L' \hat{\beta} \sim \chi_c^2$$

has χ_c^2 -distribution, and is called the Wald test statistic of the hypothesis H_0 .

4.5.2 Wald confidence intervals

Expression (4.10) can also be used to derive (approximate) confidence limits for $L'\beta$, when L is a single contrast, that is, L is a $p \times 1$ vector. Then a $(1 - \alpha)$ confidence

interval for $L'\beta$ is given by

$$L'\hat{\beta} \pm z_{1-\alpha/2} \cdot \sqrt{L'(X'WX)^{-1}L}$$

The above confidence interval is called the Wald confidence interval for $L'\beta$.

4.5.3 Example: Wald statistic and confidence interval

We illustrate hypothesis testing using Wald statistics, and calculation of Wald confidence intervals, by again using the probit analysis example treated in more detail in Section 5.3.4. Table 5.1 in Section 5.3.4 shows the number of beetles dead after 5 hours of exposure to carbon disulphide gas at various concentrations (data from Bliss, 1935). We fit a probit model with $x_i = \log(\text{concentration}_i)$. Now we want to test whether Dose (actually: logdose) is a significant predictor in the model using a Wald test. Furthermore, we calculate a Wald confidence interval for the regression slope.

We use the following SAS code:

```
PROC GENMOD DATA=beetles;
  TITLE 'Wald Tests and Confidence Intervals';
  MODEL y/n = logdose / DIST=binomial LINK=probit TYPE3 WALD WALDCI;
RUN;
```

The Wald test for effects in the model is requested by the options `TYPE3 WALD`; `TYPE3` alone would produce likelihood ratio tests (see Sections 4.6 to 4.8). The Wald confidence intervals for model parameters are requested through the `WALDCI` option.

Output for the above SAS code is included with Appendix 4.1.

4.6 Log-likelihood Ratio Statistic

One way of assessing the fit of a model of interest is to compare it with a more general model that has the maximum number of parameters that can be estimated. This latter model is called the **saturated model**. The saturated model is a generalized linear model with the same distribution and link function as the model of interest.

If there are n observations y_i , $i = 1, \dots, n$, all with potentially different values for the linear component $x_i'\beta$, then a saturated model can be specified with n parameters. This model is also called the **maximal** or **full model**.

If some of the observations have the same linear component or covariate pattern, for example, they correspond to the same combination of factor levels and have the same values of continuous explanatory variables, they are called **replicates**. In this case, the maximum number of parameters that can be estimated for the saturated model is equal to the number of potentially different linear components, which may be less than n .

In general, let m denote the maximum number of parameters that can be estimated. Let $\beta_{\max}$ denote the parameter vector of the saturated model and $b_{\max}$ denote the ML estimator of the $\beta_{\max}$. The likelihood function of the saturated model evaluated at $b_{\max}$, which is $L(b_{\max}; y)$, will be larger than any other likelihood function for these observations (with the same assumed distribution and link function).

Now let $L(b; y)$ denote the maximum value of the likelihood function of the model of interest. Then the likelihood ratio

$$\lambda = \frac{L(b_{\max}; y)}{L(b; y)}$$

provides a way of assessing the goodness of fit of the model. In practice, the logarithm of the likelihood ratio, which is the difference of the log-likelihood functions is used:

$$\log(\lambda) = l(b_{\max}; y) - l(b; y)$$

Large values of $\log(\lambda)$ suggest that the model of interest is a poor description of the data relative to the saturated model. To determine the critical region for $\log(\lambda)$ we need its sampling distribution.

In the next section we see that $2\log(\lambda)$ has a chi-squared distribution. Therefore $2\log(\lambda)$, rather than $\log(\lambda)$, is the more commonly used statistic. This statistic was called the **deviance** by Nelder and Wedderburn (1972).

4.7 Sampling Distribution of the Deviance

The deviance, also called the **log-likelihood ratio statistic**, is

$$D = 2[l(b_{\max}; y) - l(b; y)]$$

From Equation (4.4), if b is the ML estimator for β (so that $U(b) = 0$), we have

$$l(\beta) - l(b) = -\frac{1}{2}(\beta - b)' \mathbf{I}(b)(\beta - b)$$

approximately. Therefore the statistic

$$2[l(b; y) - l(\beta; y)] = (\beta - b)' \mathbf{I}(b)(\beta - b)$$

has, from (4.7), a χ_p^2 distribution, where p is the number of parameters in β .

From this result the sampling distribution of the deviance can be derived:

$$\begin{aligned} D &= 2[l(b_{\max}; y) - l(b; y)] \\ &= 2[l(b_{\max}; y) - l(\beta_{\max}; y) - l(b; y) + l(\beta; y) + l(\beta_{\max}; y) - l(\beta; y)] \\ &= 2[l(b_{\max}; y) - l(\beta_{\max}; y)] \\ &\quad - 2[l(b; y) - l(\beta; y)] + 2[l(\beta_{\max}; y) - l(\beta; y)] \end{aligned} \quad (4.11)$$

The first term in (4.11) has a χ_m^2 distribution, where m is the number of parameters in the saturated model. The second term has a χ_p^2 distribution, where p is the number of parameters in the model of interest. The third term, $v = 2[l(\beta_{\max}; y) - l(\beta; y)]$, is a positive constant which will be near zero if the model of interest fits the data almost as well as the saturated model fits. Therefore, the sampling distribution of the deviance is approximately

$$D \sim \chi_{m-p}^2(v)$$

where v is the non-centrality parameter of the distribution. The deviance forms the basis of most hypothesis testing in generalized linear models (see Section 4.9).

When the response variables Y_i are normally distributed, then D has an exact chi-square distribution. In this case, however, D depends on $\text{Var}(Y_i) = \sigma^2$ which, in practice, is unknown. This implies that D cannot be used as a goodness of fit statistic (see Example 4.8.2).

4.7.1 Likelihood ratio tests

Alternatively to Wald tests, hypotheses on the parameters β in generalized linear models can be carried out using **likelihood ratio tests (LRT)**. For simplicity, let us assume that the parameter vector β can be partitioned as $\beta = [\beta_1, \beta_2]'$ and we want to test the following null hypothesis

$$H_0 : \beta_2 = 0 \quad \text{against} \quad H_1 : \beta_2 \neq 0$$

where β_2 represents c independent parameters.

Let $l_1 = l(\hat{\beta}; y)$ denote the maximum value of the log-likelihood function for the larger model, where the full parameter vector $\beta = [\beta_1, \beta_2]'$ is fitted, and let $l_0 = l(\tilde{\beta}; y)$ denote the maximum value of the likelihood function for the smaller model (nested in Model 1), where only the parameters β_1 are fitted (with $\beta_2 = 0$). Then twice the log-likelihood ratio for those two models is given by

$$\begin{aligned} 2 \log(\lambda) &= 2(l_1 - l_0) \\ &= 2 \{ -[l(b_{\max}) - l_1] + [l(b_{\max}) - l_0] \} \\ &= D_0 - D_1 \end{aligned}$$

where D_1 and D_0 are the deviance for Model 1 and Model 0, respectively. Under the null-hypothesis $H_0 : \beta_2 = 0$ the “Model 0 – Model 1” difference of deviances

$$D_0 - D_1 = 2(l_1 - l_0) \sim \chi_c^2$$

has approximate χ_c^2 -distribution, and is called the Likelihood Ratio Test statistic of the hypothesis H_0 .

4.7.2 Likelihood ratio based confidence intervals

Alternatively to Wald confidence intervals for a single parameter β_j , likelihood ratio based confidence intervals, namely confidence intervals based on the **profile likelihood** can be calculated.

Let $\beta = [\beta_1, \dots, \beta_p]'$ be the parameter vector of the generalized linear model, and let β_j , $1 \leq j \leq p$ be the parameter of interest. The profile log-likelihood function for

β_j is defined as

$$l^*(\beta_j) = \max_{\beta_{(j)}} l(\beta)$$

where $\beta_{(j)}$ is the vector β with the j -th element fixed at β_j , and $l(\beta) = l(\beta; y)$ is the log-likelihood function. If $l = l(\hat{\beta}) = l(\hat{\beta}; y)$ is the maximum log-likelihood (namely $l(\beta)$ evaluated at the ML estimate $\hat{\beta}$), then

$$2[l - l^*(\beta_j)] \sim \chi_1^2$$

has approximate χ_1^2 -distribution. Thus a $(1 - \alpha)$ confidence interval for β_j is given by

$$C_{1-\alpha}(\beta_j) = \left\{ \beta_j \mid 2[l - l^*(\beta_j)] \leq \chi_{1;1-\alpha}^2 \right\}$$

4.7.3 Example: LRT statistic and confidence interval

We illustrate hypothesis testing using likelihood ratio statistics, and calculation of profile likelihood based confidence intervals, by again using the probit analysis example.

We use the following SAS code:

```
PROC GENMOD DATA=beetles;
  TITLE 'Likelihood Ratio Tests and Confidence Intervals';
  MODEL y/n = logdose / DIST=binomial LINK=probit TYPE3 LRCI;
RUN;
```

The likelihood ratio test for effects in the model is requested by the TYPE3 option in the MODEL statement (the likelihood ratio test is the SAS default). Profile likelihood confidence intervals for model parameters are requested through the LRCI option.

Output for the above SAS code is included with Appendix 4.2.

4.8 Examples: Deviance

4.8.1 Deviance for a Binomial model

If $Y_i \sim \text{Bin}(\pi_i, n_i)$ are independent, the log-likelihood function is

$$l(\pi; y) = \sum_{i=1}^n \left[y_i \log(\pi_i) + (n_i - y_i) \log(1 - \pi_i) + \log \binom{n_i}{y_i} \right]$$

For a saturated model, the π_i 's are all different, so $\beta_{\max} = (\pi_1, \dots, \pi_n)'$. The ML estimates of the π_i are $\hat{\pi}_i = y_i/n_i$, so the maximum of the log-likelihood function is

$$l(b_{\max}; y) = \sum_{i=1}^n \left[y_i \log \left(\frac{y_i}{n_i} \right) + (n_i - y_i) \log \left(\frac{n_i - y_i}{n_i} \right) + \log \left(\frac{n_i}{y_i} \right) \right]$$

For any other model with $p < n$ parameters, let $\hat{\pi}_i$ denote the ML estimates of the probabilities, and $\hat{y}_i = n_i \hat{\pi}_i$ denote the fitted values. Then the log-likelihood evaluated at these values is

$$l(b; y) = \sum_{i=1}^n \left[y_i \log \left(\frac{\hat{y}_i}{n_i} \right) + (n_i - y_i) \log \left(\frac{n_i - \hat{y}_i}{n_i} \right) + \log \left(\frac{n_i}{y_i} \right) \right]$$

Therefore the deviance is

$$\begin{aligned} D &= 2[l(b_{\max}; y) - l(b; y)] \\ &= 2 \sum_{i=1}^n \left[y_i \log \left(\frac{y_i}{\hat{y}_i} \right) + (n_i - y_i) \log \left(\frac{n_i - y_i}{n_i - \hat{y}_i} \right) \right] \end{aligned}$$

4.8.2 Deviance for a Normal linear model

For independent $Y_i \sim N(\mu_i, \sigma^2)$ we consider the general linear model

$$E(Y_i) = \mu_i = x_i' \beta; \quad i = 1, \dots, n$$

The log-likelihood function is

$$l(\beta, y) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu_i)^2 - n \log(\sigma \sqrt{2\pi})$$

For a saturated model all the μ_i are different, so β has n elements $\beta_{\max} = (\mu_1, \dots, \mu_n)'$. The ML estimates of these parameters are $\hat{\mu}_i = y_i$. Therefore the maximum value of the log-likelihood function is

$$l(b_{\max}; y) = -n \log(\sigma \sqrt{2\pi})$$

For any other model with $p < n$ parameters, let

$$b = (X'X)^{-1}X'y$$

be the ML estimator for β . The corresponding maximum value of the log-likelihood function is

$$l(b, y) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - x_i'b)^2 - n \log(\sigma \sqrt{2\pi})$$

Therefore the deviance is

$$\begin{aligned}
 D &= 2[l(b_{\max}; y) - l(b; y)] \\
 &= \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - x_i' b)^2 \\
 &= \frac{1}{\sigma^2} (y - Xb)'(y - Xb)
 \end{aligned}$$

where the design matrix X has rows x_i' . The term $(y - Xb)$ can be written as

$$\begin{aligned}
 y - Xb &= y - X(X'X)^{-1}X'y \\
 &= [I - X(X'X)^{-1}X']y \\
 &= [I - H]y
 \end{aligned}$$

where $H = X(X'X)^{-1}X'$ is the so-called **hat matrix**. Therefore the quadratic form in D can be written as

$$\begin{aligned}
 (y - Xb)'(y - Xb) &= [(I - H)'y]'[(I - H)y] \\
 &= y'(I - H)y
 \end{aligned}$$

because $H' = H$ and $HH = H$. The rank of I is n , and the rank of H is $n - p$, so $D = y'(I - H)y/\sigma^2$ has an exact chi-square distribution with $n - p$ degrees of freedom. (The noncentrality parameter of the distribution is $v = (X\beta)'(I - H)(X\beta)/\sigma^2 = 0$.)

The term **scaled deviance** is sometimes used for

$$\sigma^2 D = \sum_{i=1}^n (y_i - x_i' b)^2$$

If the model fits well, then $D \sim \chi_{n-p}^2$. The expected value for a χ_{n-p}^2 random variable is $n - p$, so the expected value of D is $n - p$. This provides an estimate for σ^2 as

$$\hat{\sigma}^2 = \sum_{i=1}^n (y_i - x_i' b)^2 / (n - p)$$

Some statistical programs such as SAS output the scaled deviance for a Normal linear model and call $\hat{\sigma}$ the scale parameter.

4.8.3 Deviance for a Poisson model

If $Y_i \sim \text{Poisson}(\lambda_i)$ are independent, the log-likelihood function is

$$l(\beta; y) = \sum_{i=1}^n y_i \log(\lambda_i) - \sum_{i=1}^n \lambda_i - \sum_{i=1}^n \log(y_i!)$$

For a saturated model all the λ_i are different so β has n elements $\beta_{\max} = (\lambda_1, \dots, \lambda_n)'$.

The ML estimates of these parameters are $\hat{\lambda}_i = y_i$. Therefore the maximum value of the log-likelihood function is

$$l(b_{\max}; y) = \sum_{i=1}^n y_i \log(y_i) - \sum_{i=1}^n y_i - \sum_{i=1}^n \log(y_i!)$$

For any other model of interest with $p < n$ parameters, let b denote the ML estimate for β from which the estimates $\hat{\lambda}_i$ can be calculated. The fitted values are then $\hat{y}_i = \hat{\lambda}_i$ since $E(Y_i) = \lambda_i$. The maximum value of the log-likelihood in this case is

$$l(b; y) = \sum_{i=1}^n y_i \log(\hat{y}_i) - \sum_{i=1}^n \hat{y}_i - \sum_{i=1}^n \log(y_i!)$$

Therefore the deviance is

$$\begin{aligned} D &= 2[l(b_{\max}; y) - l(b; y)] \\ &= 2 \left[\sum_{i=1}^n y_i \log \left(\frac{y_i}{\hat{y}_i} \right) - \sum_{i=1}^n (y_i - \hat{y}_i) \right] \end{aligned}$$

For most models it can be shown that $\sum y_i = \sum \hat{y}_i$. Therefore D can be written as

$$\begin{aligned} D &= 2 \sum_{i=1}^n y_i \log \left(\frac{y_i}{\hat{y}_i} \right) \\ &= 2 \sum_{i=1}^n O_i \log \left(\frac{O_i}{E_i} \right) \end{aligned}$$

where O_i and E_i respectively denote the observed and expected values of Y_i .

In the Poisson case, the value of D can be calculated from data and does not depend, as in the Normal case, on an unknown parameter σ^2 . The calculated value for D can be compared to the χ^2_{n-p} distribution to judge goodness of fit.

4.8.4 Example: Deviance in Poisson Regression

Table 4.1 shows the number of cases of AIDS in Australia by date of diagnosis for 20 successive quarters from 1984 to 1988.

Table 4.1: AIDS Cases

Year	Quarter			
	1	2	3	4
1984	1	6	21	41
1985	27	39	31	30
1986	43	51	63	70
1987	88	97	91	104
1988	110	113	149	159

In this early phase of the epidemic, the numbers seemed to increase exponentially. Therefore, a possible model is the Poisson distribution with parameter $\lambda_i = i^\beta$, where $i = 1, \dots, 20$ denotes the quarter in question. Equivalently, the model is

$$\log(\lambda_i) = \log(i) \cdot \beta = x_i \beta$$

where $x_i = \log(i)$ is the regressor variable. A slightly more general model is

$$g(\lambda_i) = \log(\lambda_i) = \beta_1 + x_i \beta_2$$

The former model is the simple linear regression model without an intercept, whereas the latter model is simple linear regression. For the latter Poisson regression model, therefore, the three components of the generalized linear model are as follows:

1. **Error Structure / Distribution:** Independent response variables

$$Y_i \sim \text{Poisson}(\lambda_i).$$

2. **Linear Predictor:** Parameters β and explanatory variables X such that $\eta_i = \beta_1 + x_i \beta_2$, or, in matrix notation

$$\eta = X\beta$$

where $x_i = \log(i)$, and X and $\beta = (\beta_1, \beta_2)'$ are the design matrix and parameter vector for a simple linear regression.

3. **Link Function:** The link function g is the logarithm, since $g(\lambda_i) = \log(\lambda_i) = \beta_1 + x_i\beta_2$. In matrix notation:

$$g(\lambda) = \log(\lambda) = X\beta$$

The SAS code to fit the above model is as follows:

```
PROC GENMOD DATA=aids;  
  TITLE 'Deviance in Poisson Regression';  
  MODEL count = logquarter / DIST=poisson LINK=log p;  
RUN;
```

The output is included with Appendix 4.3.

We note that the deviance for the simple regression model is $D = 21.76$, which is reasonably small for a variate with χ^2_{18} distribution (the degrees of freedom are $m - p = 20 - 2 = 18$). We conclude that the log-linear model fits the data reasonably well. If we drop the intercept from the model, however, which can be achieved by the `NOINT` option in the `MODEL` statement, the deviance is $D = 52.38$ with $20 - 1 = 19$ degrees of freedom. Thus, the regression model without intercept does not fit the data well.

Chapter 5

Binary Data and Logistic Regression

5.1 Probability Distributions

In this chapter we consider generalized linear models for binary data. For example, the responses might be alive or dead, present or absent, “success” or “failure”.

We define the **binary random variable** as

$$Z = \begin{cases} 1 & \text{if the outcome is a “success”} \\ 0 & \text{if the outcome is a “failure”} \end{cases}$$

with probabilities $\text{Prob}(Z = 1) = \pi$ and $\text{Prob}(Z = 0) = 1 - \pi$. The random variable Z has the Binomial $\text{Bin}(\pi, 1)$ distribution, and we write $Z \sim \text{Bin}(\pi, 1)$.

If there are n such random variables $Z_1, \dots, Z_n$ which are independent, with $\text{Prob}(Z_j = 1) = \pi_j$, then their joint probability is

$$\prod_{j=1}^n \pi_j^{z_j} (1 - \pi_j)^{1-z_j} = \exp \left[\sum_{j=1}^n z_j \log \left(\frac{\pi_j}{1 - \pi_j} \right) + \sum_{j=1}^n \log(1 - \pi_j) \right] \quad (5.1)$$

This distribution, namely the product binomial distribution, is a member of the exponential family (see Section 2.2.3).

When all the π_j 's are equal, we can define

$$Y = \sum_{j=1}^n Z_j$$

so that Y represents the number of successes in n “trials”. The random variable Y has the $\text{Bin}(\pi, n)$ distribution which has probability function (pf)

$$\text{Prob}(Y = y) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}, \quad y = 0, 1, \dots, n \quad (5.2)$$

Finally, we consider the general case of N independent random variables $Y_1, \dots, Y_N$ corresponding to the number of successes in N different strata or groups of size n_i (see Table 5.0 below).

Table 5.0: Frequencies for N Binomial Distributions

Group	Successes	Failures	Total
	Y_i	$n_i - Y_i$	n_i
1	y_1	$n_1 - y_1$	n_1
2	y_2	$n_2 - y_2$	n_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	y_N	$n_N - y_N$	n_N

If $Y_i \sim \text{Bin}(\pi_i, n_i)$, $i = 1, \dots, N$, the log-likelihood function is

$$l(\pi; y) = \sum_{i=1}^N \left[y_i \log \left(\frac{\pi_i}{1 - \pi_i} \right) + n_i \log(1 - \pi_i) + \log \binom{n_i}{y_i} \right] \quad (5.3)$$

where $y = (y_1, \dots, y_N)'$.

5.2 Generalized Linear Models for Binary Data

We want to model the probability of success $\pi_1, \dots, \pi_N$ in the N groups, in terms of explanatory variables which characterize the subgroup. Since $E(Y_i) = n_i \pi_i$ we model the probabilities π_i as

$$g(\pi_i) = x_i' \beta$$

where x_i is a vector of explanatory variables, β is a vector of parameters and g is the link function.

The most simple case is the **linear model**

$$g(\pi_i) = \pi_i = x_i' \beta$$

where g is the identity link function. This model is occasionally used in practice, but it has the disadvantage that although all π_i are probabilities between zero and 1, the fitted values $\hat{\pi}_i = x_i' \hat{\beta}$ might be smaller than 0 or larger than 1.

To ensure that estimates of probabilities are restricted to the interval $[0, 1]$, π is often modelled using a cumulative distribution function (cdf) such that

$$\pi = \int_{-\infty}^t f(s) ds$$

where $f(s)$ is a pdf which is called the **tolerance distribution**. The two most commonly used examples of tolerance distributions are considered in Sections 5.3.1 and 5.3.2 below.

5.3 Dose Response Models

5.3.1 Probit model

Historically one of the first uses of regression-like models for binomial data was for so-called bioassay results. Responses were the proportions or percentages of “successes”, for example, the proportion of animals killed by various dose levels of a toxic substance. Such data are sometimes called **quantal responses**. The aim is to describe the probability of “success” (here: death) π_i as a function of the dose x_i . For example,

$$g(\pi_i) = \beta_1 + x_i \beta_2$$

One of the original models used for bioassay data is called the **probit model**, where the Normal distribution is used as the tolerance distribution.

One way to motivate the concept of tolerance distribution in general, and the probit model in particular, is as follows: Let us assume that each experimental animal has a **tolerance** T to the toxic substance (the value of T itself is not observed): if the tolerance is smaller than the dose, $T < x_i$, the animal dies, otherwise, if $T \geq x_i$, the

animal survives. Furthermore, it is assumed that the tolerance T is a random variable with a distribution in the population of experimental animals.

If we choose the Normal $N(\mu, \sigma^2)$ distribution as tolerance distribution we obtain the probit model: the probability π of an animal dying, when receiving the dose x_i , is given by

$$\begin{aligned}\pi_i &= \text{Prob}(T < x_i) \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x_i} \exp\left[-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2\right] dt \\ &= \Phi\left(\frac{x_i - \mu}{\sigma}\right)\end{aligned}$$

where Φ is the cdf of the normal $N(0, 1)$ distribution. Then we can write the probit model as

$$\begin{aligned}\Phi^{-1}(\pi_i) &= \frac{x_i - \mu}{\sigma} \\ &= -\frac{\mu}{\sigma} + x_i \frac{1}{\sigma} \\ &= \beta_1 + x_i \beta_2\end{aligned}$$

Here $\beta_1 = -\mu/\sigma$ and $\beta_2 = 1/\sigma$, and the link function is Φ^{-1} , the inverse of the cdf of the standard Normal distribution.

In bioassay and other applications of the probit model, $x = \mu$ is called the **median lethal dose** LD(50), because at this dose 50% of animals are expected to be killed. Since $\mu = -\beta_1/\beta_2$, the ML estimate of LD(50) is given by $\hat{\mu} = -\hat{\beta}_1/\hat{\beta}_2$.

5.3.2 Logistic model

Another model for binary dose-response data, which gives results usually very similar to those from the probit model, is the **logistic** or **logit model**. In this case, the tolerance distribution is the logistic distribution with pdf

$$f(t) = \frac{\beta_2 \exp(\beta_1 + t\beta_2)}{[1 + \exp(\beta_1 + t\beta_2)]^2}$$

Therefore

$$\pi = \int_{-\infty}^x f(t) dt = \frac{\exp(\beta_1 + x\beta_2)}{1 + \exp(\beta_1 + x\beta_2)}$$

This gives the link function

$$\log\left(\frac{\pi}{1-\pi}\right) = \beta_1 + x\beta_2$$

The term $\log[\pi/(1-\pi)]$ is called the **logit** of π and can be interpreted as the logarithm of the **odds**, namely $\pi/(1-\pi)$, of π . The logistic model is widely used for binomial data.

5.3.3 Complementary log log model

Several other models are also used for dose response data. For example, the **extreme value distribution** with pdf

$$f(t) = \beta_2 \exp[(\beta_1 + t\beta_2) - \exp(\beta_1 + t\beta_2)]$$

can be used as the tolerance distribution. Then

$$\pi = \int_{-\infty}^x f(t) dt = 1 - \exp[-\exp(\beta_1 + x\beta_2)]$$

and the link function is given by the **complementary log log function**

$$g(\pi) = \log[-\log(1-\pi)] = \beta_1 + x\beta_2$$

5.3.4 Example: Dose Response Models for Binary Data

Table 5.1 shows the number of beetles dead after 5 hours of exposure to carbon disulphide gas at various concentrations (data from Bliss, 1935).

Table 5.1: Beetle Mortality

Log Dose	Number Beetles	Number killed
x_i	n_i	y_i
1.6907	59	6
1.7242	60	13
1.7552	62	18
1.7842	56	28
1.8113	63	52
1.8369	59	53
1.8610	62	61
1.8839	60	60

We first fit the logistic model

$$\pi_i = \frac{\exp(\beta_1 + x_i\beta_2)}{1 + \exp(\beta_1 + x_i\beta_2)}$$

Thus

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_1 + x_i\beta_2$$

This model is the simple linear regression model for the logit of the probabilities π_i .

Therefore, three components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_i \sim \text{Bin}(\pi_i, n_i)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = \beta_1 + x_i\beta_2$, or, in matrix notation

$$\eta = X\beta$$

where $x_i = \log(\text{Dose}_i)$, and X and $\beta = (\beta_1, \beta_2)'$ are the design matrix and parameter vector for a simple linear regression.

3. Link Function:

The link function g is the logit, since $g(\pi_i) = \log(\pi_i/(1 - \pi_i)) = \beta_1 + x_i\beta$. In matrix notation:

$$g(\pi) = \log\left(\frac{\pi}{1 - \pi}\right) = X\beta$$

The SAS code to fit the above model is as follows:

```
PROC GENMOD DATA=beetles;
  TITLE 'Logistic Model for Beetle Mortality';
  MODEL y/n = logdose / DIST=binomial LINK=logit P;
RUN;
```

In the model statement the response is specified by the statement y/n , that is, we analyse the number killed, y , out of the number exposed, n . Together with the `dist=binomial` option this code is interpreted in such a way that the distribution of y_i is taken to be $\text{Bin}(\pi_i, n_i)$.

The output is included with Appendix 5.1.

Alternatively to the logistic model, the probit and the extreme value model can also be fitted. Only the third component of the generalized linear model, namely the link function, changes from the above specifications: for the probit model, we have the probit link, and for the extreme value model we have the complementary log log link. The corresponding SAS code (MODEL statement) for the probit model is

```
MODEL y/n = logdose / DIST=binomial LINK=probit P;
```

and for the extreme value model is

```
MODEL y/n = logdose / DIST=binomial LINK=cloglog P;
```

Otherwise the SAS code remains unchanged.

Table 5.2 shows (again) the observed number of beetles killed for each dose, and the fitted values associated with the three alternative models, logistic, probit and extreme value. Finally, the deviance D associated with each model is given:

Table 5.2: Model Fit for Beetle Data

Observed Value	Fitted Value: Model		
	Logistic	Probit	Extreme Value
6	3.46	3.36	5.59
13	9.84	10.72	11.28
18	22.45	23.48	20.95
28	33.90	33.82	30.37
52	50.10	49.62	47.78
53	53.29	53.32	54.14
61	59.22	59.66	61.11
60	58.74	59.23	59.95
D	11.23	10.12	3.45

We note that the extreme value model seems to fit the data best: For most doses, the fitted values for this model are closest to the observed values, and the deviance for the extreme value model is the smallest. The deviance has $N - 2 = 6$ degrees of freedom, and the deviance for the logistic and probit models is appreciably larger than 6.

5.4 General Logistic Regression

The simple linear logistic regression model $g(\pi_i) = \log(\pi_i/(1 - \pi_i)) = \beta_1 + x_i\beta$ used in the example above is a special case of the general logistic regression model

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = x_i'\beta$$

where x_i is the i -th row of the design matrix of the model, and the entries of x_i might stand for continuous covariates and dummy variables of categorical factors. Thus the simple logistic regression model relates to the general logistic regression model much in the same way as the simple linear regression model (see Equations (1.4) and (1.12)) relates to the general linear model (1.8)–(1.11). The general logistic regression model is widely used for analyzing data involving binary or binomial responses.

5.4.1 Example: General logistic regression

Table 5.3 shows the numbers of embryogenic anthers y_{jk} of the plant species *Datura innoxia* obtained when n_{jk} anthers were prepared under different conditions (Sangwan-Norrell, 1977). One categorical factor is Storage condition, namely storage at 3 degrees Celsius (Treatment) or a control storage condition (Control). The continuous explanatory variable represents three values of centrifuging force. Aim of the analysis is to compare Treatment to Control after adjustment for Centrifuging Force.

Table 5.3: Embryogenic Anther Data

		Centrifuging Force		
		40	150	350
Control	y_{1k}	55	52	57
	n_{1k}	102	99	108
Treatment	y_{2k}	55	50	50
	n_{2k}	76	81	90

We compare three logistic models for π_{jk} , the probability of the anthers being embryogenic, where $j = 1$ for the control group and $j = 2$ for the treatment group. Furthermore $x_1 = \log(40)$, $x_2 = \log(150)$ and $x_3 = \log(350)$:

Model 1: $\text{logit}(\pi_{jk}) = \alpha_j + x_k\beta_j$ different intercepts and slopes for treatments

Model 2: $\text{logit}(\pi_{jk}) = \alpha_j + x_k\beta$ different intercepts but same slope

Model 3: $\text{logit}(\pi_{jk}) = \alpha + x_k\beta$ same intercept and same slope

All three models are general linear models for the logit of the probabilities π_i . Therefore, the three components of the generalized linear model are as follows:

1. Error structure / Distribution:

Independent response variables $Y_{jk} \sim \text{Bin}(\pi_{jk}, n_{jk})$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_{jk} = \alpha_j + x_k\beta_j$ (for Model 1), or, in matrix notation

$$\eta = X\beta$$

where $x_{jk} = \log(\text{centrifuging force})$, and X and $\beta = (\alpha_1, \alpha_2, \beta_1, \beta_2)'$ are the design matrix and parameter vector for an Analysis of Covariance Model with non-parallel slopes.

3. Link Function:

The link function g is the logit, since $g(\pi_{jk}) = \log(\pi_{jk}/(1 - \pi_{jk})) = \alpha_j + x_k\beta_j$. In matrix notation:

$$g(\pi) = \log\left(\frac{\pi}{1 - \pi}\right) = X\beta$$

The SAS code to fit Model 1 is as follows:

```
PROC GENMOD DATA=anthers;
  TITLE 'General Logistic Model for Embryogenic Anthers';
  CLASS storage;
  MODEL y/n = storage log_centerforce storage*log_centerforce
           / DIST=binomial LINK=logit P;
RUN;
```

The SAS code (MODEL statement) for fitting the simpler Model 2 is

```
MODEL y/n = storage log_centerforce / DIST=binomial LINK=logit P;
```

and for the most simple Model 3:

```
MODEL y/n = log_centerforce / DIST=binomial LINK=logit P;
```

Otherwise the SAS code remains unchanged.

The analysis results are summarized in Table 5.4:

Table 5.4: Estimates and Deviances for Embryogenic Anther Data

Parameter	ML Estimate (SE)					
	Model 1		Model 2		Model 3	
α_1	0.234	(0.628)	0.877	(0.487)	1.021	(0.481)
α_2	2.211	(0.775)	1.284	(0.498)		
β_1	-0.023	(0.127)	-0.155	(0.097)	-0.148	(0.096)
β_2	-0.341	(0.153)				
$D(df)$	0.028	2	2.619	3	8.092	4
$\Delta D(df)$			2.591	1	5.473	1

To test the null hypothesis that the slope is the same for the two treatments, we use the difference in deviances between Model 2 and Model 1, namely

$$\Delta D = D_2 - D_1 \sim \chi_1^2$$

The value of the statistic is 2.591, which corresponds to a P-value of 0.11 (χ_1^2 -distribution).

Comparison of the deviances from Models 3 and 2 provides a test for the equality of intercepts (storage conditions). The statistic is $D_3 - D_2 = 5.473$ which corresponds to a significant P-value of 0.019 (χ_1^2 -distribution) and therefore suggests that storage conditions have an important effect.

Statistics to test for the significance of model factors based on the differences in deviances, as used above, are automatically calculated by SAS PROC GENMOD if one specifies the TYPE3 option in the MODEL statement. The MODEL statement for Model 1 then looks as follows:

```
MODEL y/n = storage log_centerforce storage*log_centerforce
      / DIST=binomial LINK=logit P TYPE3;
```

The output looks very similar to an ANOVA table for a general linear model, see Appendix 5.2.

5.5 Goodness of Fit Statistics

Instead of using maximum likelihood, we could estimate the parameters, and fit the model, by minimizing the weighted sum of squares

$$SS_w = \sum_{i=1}^n \frac{(y_i - n_i\pi_i)^2}{n_i\pi_i(1 - \pi_i)}$$

since $E(Y_i) = n_i\pi_i$ and $\text{Var}(Y_i) = n_i\pi_i(1 - \pi_i)$.

This is equivalent to minimizing

$$X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where O_i represents the observed frequencies, E_i the expected frequencies, and summation is over all $2 \times N$ cells of Table 5.0 (both successes and failures). The reason is that

$$\begin{aligned} X^2 &= \sum_{i=1}^n \frac{(y_i - n_i\pi_i)^2}{n_i\pi_i} + \sum_{i=1}^n \frac{[(n_i - y_i) - n_i(1 - \pi_i)]^2}{n_i(1 - \pi_i)} \\ &= \sum_{i=1}^n \frac{(y_i - n_i\pi_i)^2}{n_i\pi_i} + \sum_{i=1}^n \frac{(-y_i + n_i\pi_i)^2}{n_i(1 - \pi_i)} \\ &= \sum_{i=1}^n \frac{(y_i - n_i\pi_i)^2(1 - \pi_i) + (y_i - n_i\pi_i)^2\pi_i}{n_i\pi_i(1 - \pi_i)} \\ &= \sum_{i=1}^n \frac{(y_i - n_i\pi_i)^2(1 - \pi_i + \pi_i)}{n_i\pi_i(1 - \pi_i)} \\ &= SS_w \end{aligned}$$

When X^2 is evaluated at the estimated expected frequencies, the statistic is

$$X^2 = \sum_{i=1}^n \frac{(y_i - n_i\hat{\pi}_i)^2}{n_i\hat{\pi}_i(1 - \hat{\pi}_i)} \quad (5.4)$$

which is called the **Pearson chi-square statistic**. This statistic is asymptotically equivalent to the deviance

$$\begin{aligned} D &= 2 \sum_{i=1}^n \left[y_i \log \left(\frac{y_i}{\hat{y}_i} \right) + (n_i - y_i) \log \left(\frac{n_i - y_i}{n_i - \hat{y}_i} \right) \right] \\ &= 2 \sum_{i=1}^n \left[y_i \log \left(\frac{y_i}{n_i\hat{\pi}_i} \right) + (n_i - y_i) \log \left(\frac{n_i - y_i}{n_i - n_i\hat{\pi}_i} \right) \right] \\ &= \sum O_i \log \left(\frac{O_i}{E_i} \right) \end{aligned} \quad (5.5)$$

The proof of the relationship between X^2 and D uses a Taylor series expansion of the function $f(O) = O \log(O/E)$ about E .

5.6 Residuals

For logistic regression there are two main forms of residuals corresponding to the goodness of fit measures D (the deviance) and X^2 (Pearson chi-square). If there are m different covariate patterns, then m different residuals can be calculated. Let Y_k denote the number of successes, n_k the number of trials, and $\hat{\pi}_k$ the estimated probability of success for the k -th covariate pattern.

The **Pearson** or **chi-squared** residual is

$$X_k = \frac{(y_k - n_k \hat{\pi}_k)}{\sqrt{n_k \hat{\pi}_k (1 - \hat{\pi}_k)}}, \quad k = 1, \dots, m \quad (5.6)$$

From (5.4) we have $\sum_{i=1}^m X_k^2 = X^2$, the Pearson chi-squared goodness of fit statistic.

The **standardized Pearson residuals** are

$$r_{Pk} = \frac{X_k}{\sqrt{1 - h_k}}, \quad k = 1, \dots, m$$

where h_k is the **leverage**, obtained as the k -th element on the diagonal of the **Hat Matrix**

$$H = W^{1/2} X (X' W X)^{-1} X' W^{1/2}$$

The **deviance residuals** can be defined similarly,

$$d_k = \text{sign}(y_k - n_k \hat{\pi}_k) \left\{ 2 \left[y_k \log \left(\frac{y_k}{n_k \hat{\pi}_k} \right) + (n_k - y_k) \log \left(\frac{n_k - y_k}{n_k - n_k \hat{\pi}_k} \right) \right] \right\}^{1/2} \quad (5.7)$$

where the term $\text{sign}(y_k - n_k \hat{\pi}_k)$ ensures that d_k has the same sign as X_k . From (5.5) we have $\sum_{i=1}^m d_k^2 = D$, the deviance. Standardized deviance residuals are defined as

$$r_{Dk} = \frac{d_k}{\sqrt{1 - h_k}}, \quad k = 1, \dots, m$$

The residuals can be used to check the adequacy of a model, similarly to their use in linear regression. For example, they can be plotted against each explanatory variable in the model to check if the assumption of linearity is valid. Normal probability plots could also be used.

5.7 Over-Dispersion

By the term “over-dispersion” in binomial data we mean that the variance of the response Y exceeds the nominal variance – in this case, the nominal binomial variance $M\pi(1 - \pi)$ if we assume $Y \sim \text{Bin}(\pi, M)$. Over-dispersion frequently occurs in practice. Unless there are good external reasons for relying on the binomial assumption it is wise to be cautious and to assume that over-dispersion is present to some extent until it is shown to be absent. Thus we should always check for overdispersion in the analysis of binomial data, by comparing the deviance or Pearson chi-square goodness of fit statistics to their asymptotic distribution (chi-squared) under the assumption of a binomial distribution of the data. The expected value of a chi-square variable with $N - p$ degrees of freedom is $N - p$. Thus a good rule of thumb is that the deviance and Pearson chi-square goodness of fit statistics should be in the order of magnitude of their degrees of freedom.

Over-dispersion can arise in a number of ways. The most simple and perhaps most common mechanism is clustering in the population. Families, households, litters, colonies and neighbourhoods are common instances of clusters in populations. Clusters usually vary in size, but for simplicity we assume that the cluster size, k , is fixed and that the M individuals sampled come from $N = M/k$ clusters. In the i -th cluster the number of positive respondents, Y_i , is assumed to have the binomial distribution $\text{Bin}(\pi_i, k)$, where the parameter π_i varies from cluster to cluster. Thus the total number of positive respondents is

$$Y = Y_1 + Y_2 + \cdots + Y_N$$

If we write $E(\pi_i) = \pi$ and $\text{Var}(\pi_i) = \tau^2\pi(1 - \pi)$ it can be shown that the unconditional mean and variance of Y are

$$\begin{aligned} E(Y) &= M\pi \\ \text{Var}(Y) &= M\pi(1 - \pi) [1 + (k - 1)\tau^2] \\ &= \sigma^2 M\pi(1 - \pi) \end{aligned} \tag{5.8}$$

Note that the dispersion parameter $\sigma^2 = 1 + (k - 1)\tau^2$ depends on the cluster size and on the variability of π from cluster to cluster, but not on the total sample size

M . This fact is important because it allows us to proceed as if the observations were binomially distributed and to estimate the dispersion parameter from the residuals (see next section).

The Beta-Binomial distribution is sometimes used as an alternative model for over-dispersion. This distribution has the property that the variance ratio

$$\frac{\text{Var}(Y)}{M\pi(1-\pi)}$$

is a linear function of M , rather than a constant as in (5.8). McCullagh and Nelder (1989) state that examples examined by them favour the constant dispersion factor over the Beta-Binomial model.

5.7.1 Parameter estimation and inference

With specific forms of over-dispersion, such as the Beta-Binomial model, one can use maximum likelihood to estimate the regression parameters and the dispersion parameter jointly. In what follows, however, we do not assume a specific distributional model for the over-dispersion, but rather assume that the effect of over-dispersion has the form (5.8): in other words, the mean is unaffected by the over-dispersion but the variance is inflated by an unknown factor σ^2 .

With this form of over-dispersion, model fit and estimation proceed as before, that is, as if the binomial distribution did still apply. The only difference is that in Equations such as (4.7) and (4.8) (Wald statistic) the chi-square approximation χ_p^2 is replaced by $\sigma^2\chi_p^2$; thus the Wald statistic no longer is $(\hat{\beta} - \beta)'X'WX(\hat{\beta} - \beta) \sim \chi_p^2$, but

$$(\hat{\beta} - \beta)'X'WX(\hat{\beta} - \beta)/\sigma^2 \sim \chi_p^2 \quad (5.9)$$

Similarly the asymptotic sampling distribution of the regression coefficients $\hat{\beta}$ is no longer $N(\beta, \mathbf{I}^{-1}) = N(\beta, (X'WX)^{-1})$, but

$$\hat{\beta} \sim N(\beta, \sigma^2\mathbf{I}^{-1}) = N(\beta, \sigma^2(X'WX)^{-1}) \quad (5.10)$$

The remaining problem is to estimate the dispersion parameter σ^2 ; an estimate for σ^2 is needed to carry out tests on β and to calculate confidence intervals. The problem is analogous to estimating σ^2 in Normal linear models.

One estimator for σ^2 is given by

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{N-p} \sum_{i=1}^N \frac{(y_i - n_i \hat{\pi}_i)^2}{n_i \hat{\pi}_i (1 - \hat{\pi}_i)} \\ &= X^2 / (N - p)\end{aligned}\tag{5.11}$$

which is the ratio of the Pearson chi-square statistic over its degrees of freedom. Alternatively, one can calculate the estimator

$$\hat{\sigma}^2 = D / (N - p)\tag{5.12}$$

which is the ratio of the deviance over its degrees of freedom. However, if the n_i are close to one (sparse case) (5.12) is not a consistent estimator of σ^2 , while (5.11) remains a consistent estimator. For this reason, (5.11) is the preferred estimator at least in the sparse case.

5.7.2 Example: Over-dispersion

In an experiment, two medications to kill ticks infesting sheep were compared. Fifty ticks were placed on 16 sheep each, and then eight sheep each were randomized to be treated with either a new medication (Test) or a control medication (Reference). After three days, the numbers of life ticks remaining on each individual sheep were counted. The data are presented in Table 5.5.

Table 5.5: Life Ticks Remaining on Sheep

Test Treatment		Reference Treatment	
Ticks placed	Ticks remaining	Ticks placed	Ticks remaining
50	4	50	10
50	26	50	9
50	3	50	9
50	10	50	9
50	15	50	6
50	24	50	17
50	6	50	15
50	5	50	19

Let π_T be the probability that a tick is removed (killed) by the test medication, and π_R the probability that a tick is removed (killed) by the reference medication. We can therefore fit the one-way model

$$g(\pi_i) = \alpha + \alpha_i, \quad i = T, R$$

for some link function g . Traditionally, analysis results for such trials are expressed as estimates for the ratio $r = \pi_T/\pi_R$, so that we use a log-link for the generalized linear model, that is, we assume $\log(\pi_i) = \alpha + \alpha_i$. Then, given ML estimates for the parameters of the model, the ML estimate for $\log(\pi_T/\pi_R) = \alpha_T - \alpha_R$ is given by $\hat{\alpha}_T - \hat{\alpha}_R$, so that the ML estimate for r is given by

$$\hat{r} = \frac{\hat{\pi}_T}{\hat{\pi}_R} = \exp(\hat{\alpha}_T - \hat{\alpha}_R)$$

The SAS code to fit the Model to the data is as follows:

```
PROC GENMOD DATA=ticks;
  TITLE 'Logistic One-Way ANOVA: Overdispersion';
  CLASS treatment;
  MODEL y/n = treatment / DIST=binomial LINK=log SCALE=pearson;
  LSMEANS treatment / DIFF CL;
RUN;
```

The output is included with Appendix 5.3.

We notice on page 1 of the output that the both the Pearson chi-square statistic and the deviance are significantly larger than their degrees of freedom, by a factor of almost six. Thus we clearly have a case of overdispersion, due to clustering, where the 16 individual sheep form the 16 clusters. To account for over-dispersion in hypothesis tests and confidence interval calculations, we have specified in the `MODEL` statement the `SCALE=pearson` option, which causes SAS to estimate the dispersion parameter based on the Pearson chi-square statistic (see Equation (5.11)) and make the relevant adjustment to test statistics and interval estimates (in Equations (5.9) and (5.10) the unknown value of σ^2 is replaced by its estimate from (5.11)).

The estimate of the scale (dispersion) parameter on page two of the output (labeled “Scale”) is for σ , thus it is equal to the square root of the ratio of the Pearson chi-square statistic over its degrees of freedom (the latter labeled “Value/DF” by SAS).

5.8 Logistic Models and the Odds Ratio

When comparing two groups regarding the probability of success, namely π_1 for Group 1 and π_2 for Group 2, statistical inference is usually about one of three contrasts of the parameters π_1 and π_2 , namely

- the difference $RD = \pi_1 - \pi_2$, often called the **risk difference** in medical applications
- the ratio $RR = \pi_1/\pi_2$, often called the **risk ratio** in medical applications
- the odds ratio

$$\omega = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}$$

Generalized linear models for binary data with **logistic** link function are most convenient for statistical inference on the **odds ratio**. Under such models, the logit of probability of success π_i is written as a function of a vector of explanatory variables $x_i = (x_{i0}, \dots, x_{i,p-1})'$ as

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = x_i'\beta$$

where $\beta = (\beta_0, \dots, \beta_{p-1})'$.

If we now distinguish Group 1 and Group 2 by an additional binary explanatory variable such that $x_{i,p} = 1$ indicates Group 1, and $x_{i,p} = 0$ indicates Group 2, the logits or log odds of the success probabilities for Group 1 and Group 2 are respectively given by

$$\begin{aligned}\log\left(\frac{\pi_{1i}}{1 - \pi_{1i}}\right) &= x'_i\beta + 1 \cdot \beta_p, & \text{for Group 1} \\ \log\left(\frac{\pi_{2i}}{1 - \pi_{2i}}\right) &= x'_i\beta + 0 \cdot \beta_p, & \text{for Group 2}\end{aligned}$$

provided that the other explanatory variables, x_i , remain the same. The difference of the log odds of the success probabilities for Group 1 and Group 2 are therefore given by

$$\begin{aligned}\log\left(\frac{\pi_{1i}}{1 - \pi_{1i}}\right) - \log\left(\frac{\pi_{2i}}{1 - \pi_{2i}}\right) &= \log\left[\frac{\pi_{1i}/(1 - \pi_{1i})}{\pi_{2i}/(1 - \pi_{2i})}\right] \\ &= \beta_p \\ &= \log(\omega)\end{aligned}$$

In a logistic model, therefore, a parameter associated with a binary explanatory variable indicating two different groups represents the log odds ratio of success probabilities between the two groups in question. Statistical inference (point and interval estimates, hypothesis tests) about the log odds ratio can therefore easily be carried out using logistic models. Furthermore, by taking the antilog of point and interval estimates for the log odds ratio (here: the antilog of point and interval estimates for the parameter β_p associated with the indicator variable in question), we obtain point and interval estimates for the odds ratio itself.

Similarly, for a continuous variable x_{ik} , a one unit increase in x_{ik} will result in an additive effect of β_k on the log odds scale; therefore, a one unit increase in x_{ik} will result in multiplicative effect of e^{β_k} on the odds scale.

Thus, in logistic models for binomial data, contrasts (differences) of parameters associated with categorical variables and parameter estimates associated with continuous variables, are often interpreted on the exponential scale e^β , namely in terms of odds ratios.

5.8.1 Example: Odds ratio – categorical variable

We return to the experiment on two medications, Test and a control (Reference), for killing ticks (the data are presented in Table 5.5 above).

As before, let π_T be the probability that a tick is killed by the test medication, and π_R the probability that a tick is killed by the reference medication. We therefore fit the one-way model

$$g(\pi_i) = \alpha + \alpha_i, \quad i = T, R$$

for some link function g . We now want to express the analysis results as an estimate and confidence interval for the odds ratio

$$\omega = \frac{\pi_T/(1 - \pi_T)}{\pi_R/(1 - \pi_R)}$$

We use a logit link for the generalized linear model, that is, we assume

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \alpha + \alpha_i, \quad i = T, R$$

Then, given ML estimates for the parameters of the model, the ML estimate for $\log(\omega) = \alpha_T - \alpha_R$ is given by $\hat{\alpha}_T - \hat{\alpha}_R$, so that the ML estimate for ω is given by

$$\hat{\omega} = \exp(\hat{\alpha}_T - \hat{\alpha}_R)$$

The SAS code to fit the Model to the data is as follows:

```
PROC GENMOD DATA=ticks;
  TITLE 'Logistic One-Way ANOVA: Inference for Odds Ratio';
  CLASS treatment;
  MODEL y/n = treatment / DIST=binomial LINK=logit SCALE=pearson;
  LSMEANS treatment / DIFF CL;
RUN;
```

The output is included with Appendix 5.4.

The estimate of the log odds ratio is -0.033 , with a 95% Wald confidence interval of -0.1779 to 0.1713 . Taking the antilog (using a pocket calculator), the ML estimate for the odds ratio is $\hat{\omega} = 0.968$, 95% CI 0.837 to 1.187 . Thus the odds of a tick getting

killed by the test treatment are only a little lower than the odds of getting killed by the reference treatment.

(Note: For this specific type of generalized linear model, namely a **logistic model for binomial data**, there is a special SAS procedure called LOGIST. PROC LOGIST has options to automatically calculate estimates and confidence intervals for odds ratios – without intervention of a pocket calculator.)

5.8.2 Example: Odds ratio – continuous variable

We return to the data of Table 5.1, the number of beetles dead after 5 hours of exposure to carbon disulphide gas at various concentrations. We fit the logistic model

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_1 + x_i\beta_2$$

This model is the simple linear regression model for the logit of the probabilities π_i . The SAS code to fit the above model is the same as that used in Section 5.3.4 and its output is included with Appendix 5.1.

```
PROC GENMOD DATA=beetles;
  TITLE 'Simple Logistic Regression: Inference for Odds Ratio';
  MODEL y/n = logdose / DIST=binomial LINK=logit P;
RUN;
```

The ML estimate for β_2 is 32.27. Thus every unit increase in log dose increases the log odds of a beetle being killed by 32.27. To work with more manageable numbers: every 0.1 increase in log dose is associated with an increase in the log odds of a beetle being killed by 3.227 (95% CI 2.856 to 4.000). In other words again, every 0.1 increase in log dose is associated with a $\exp(3.227) = 25.2$ times increase in the odds of a beetle being killed.

Chapter 6

Poisson Regression and Log-Linear Models

6.1 Introduction

The number of times an event occurs is a common form of data. Examples of **count** or **frequency** data include the numbers of people in each cell of a contingency table that summarizes survey responses.

The **Poisson distribution** is often used to model count data. If Y is the number of occurrences, its probability distribution can be written as

$$f(y) = \frac{\mu^y e^{-\mu}}{y!}, \quad y = 0, 1, 2, \dots$$

where μ is the average number of occurrences. It can be shown that $E(Y) = \mu$ and $\text{Var}(Y) = \mu$.

The parameter μ requires careful definition. Often it needs to be described as a rate; for example, the average number of customers who buy a particular car out of every 100 customers who enter the store. For motor vehicle crashes, the rate parameter may be described in many different ways: crashes per 1000 population, crashes per 1000 licensed drivers, crashes per 1000 cars or crashes per 100,000 km traveled. The time scale should be included in the definition; for example, the crash rate is usually specified as the rate per year (e.g. crashes per 100,000 km per year).

More generally, the rate is specified in terms of units of “exposure”, often time. For

injuries or disease occurrences, each person is exposed to the risk of injury or disease for a certain time period, so the rate may be defined in terms of person-years “at risk”.

The effect of explanatory variables on the response Y is modelled through the “rate” parameter μ . The present chapter describes models for two situations.

In the first situation, the events relate to varying amounts of “exposure” which need to be taken into account when modeling the rate of events. **Poisson regression** is used in this case. The other explanatory variables (in addition to “exposure”) may be continuous or categorical.

In the second situation, “exposure” is constant (and therefore not relevant to the model) and the explanatory variables are usually categorical. If there are only a few explanatory variables the data are summarized in a cross-classified table. The response variable is the frequency or count in each cell of the table. The variables used to define the table are all treated as explanatory variables. The study design may imply that there are some constraints on the cell frequencies (for example, the totals for each row of the table may be equal) and these need to be taken into account in the modeling. The term **log-linear model**, which basically describes the role of the link function, is used for the generalized linear models appropriate for this situation.

The next section describes Poisson regression. Subsequent sections describe relationships between probability distributions for count data constrained in various ways, and the log-linear models that can be used to analyze the data.

6.2 Poisson Regression

Let $Y_1, \dots, Y_N$ be independent random variables with Y_i denoting the number of events observed from exposure n_i for the i -th covariate pattern. The expected value of Y_i can be written as

$$E(Y_i) = \mu_i = n_i \theta_i$$

For example, suppose Y_i is the number of insurance claims for a particular make of car. This will depend on the number of cars of this make that are insured, n_i , and other variables that affect θ_i , such as the age of the car and the location where they are used. The subscript i is used to indicate the different combinations of make of car,

age, location and so on.

The dependence of θ_i on the explanatory variables is usually modelled by

$$\theta_i = e^{x_i' \beta} \quad (6.1)$$

Therefore $E(Y_i)$ can be written as

$$E(Y_i) = \mu_i = n_i \theta_i = n_i e^{x_i' \beta} \quad (6.2)$$

The three components of the generalized linear model are then as follows:

1. Error structure / Distribution:

Independent response variables $Y_i \sim \text{Poisson}(\mu_i)$.

2. Linear Predictor:

Parameters β and explanatory variables X such that $\eta_i = x_i' \beta$

3. Link Function:

The link function g is the natural logarithm, since

$$g(\mu_i) = \log(\mu_i) = \log(n_i \theta_i) = \log(n_i) + x_i' \beta \quad (6.3)$$

Equation (6.3) differs slightly from the usual specification of the linear component due to the inclusion of the term $\log(n_i)$. The term is called the **offset**. It is a known constant which is easily incorporated in the estimation procedure. As usual, the terms x_i and β describe the covariate pattern and parameters, respectively.

6.2.1 Parameter estimates and rate ratios

For a binary explanatory variable denoted by an indicator variable, namely $x_{ij} = 0$ if the factor is absent and $x_{ij} = 1$ if the factor is present, the **rate ratio** RR for presence versus absence is

$$RR = \frac{E(Y_i \mid \text{present})}{E(Y_i \mid \text{absent})} = e^{1 \cdot \beta_j - 0 \cdot \beta_j} = e^{\beta_j}$$

from (6.1), provided that all the other explanatory variables remain the same. Similarly, for a continuous variable x_{ik} , a one unit increase in x_{ik} will result in a multiplicative

effect of e^{β_k} on the rate μ . Therefore, parameter estimates associated with continuous variables, or contrasts (differences) of parameters associated with categorical variables, are often interpreted on the exponential scale e^β , namely in terms of ratios of rates.

6.2.2 Hypothesis testing and confidence intervals

Hypotheses about parameters can be tested using either the score statistic (Section 4.2), the Wald statistic (Section 4.5) or the likelihood ratio statistic (Section 4.6). Confidence intervals can be estimated similarly. For example, for parameter β_j

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\text{Var}(\hat{\beta}_j)}} \sim N(0, 1) \quad (6.4)$$

approximately. Alternatively, hypothesis testing can be performed by comparing the goodness of fit of appropriately defined nested models.

6.2.3 Residuals

The fitted values are given by

$$\hat{y}_i = \hat{\mu}_i = n_i e^{x_i' \hat{\beta}}, \quad i = 1, \dots, N$$

The fitted values are often denoted by E_i because they are estimates of the expected values $E(Y_i) = \mu_i$. Since $\text{Var}(Y_i) = E(Y_i)$ for the Poisson distribution, the standard error of Y_i is estimated by $\sqrt{E_i}$, so the **Pearson residuals** are

$$r_i = \frac{O_i - E_i}{\sqrt{E_i}} \quad (6.5)$$

where $O_i = y_i$ denotes the observed value of Y_i . As outlined in Section 5.6, these residuals can be further refined to the standardized residuals

$$r_{pi} = \frac{O_i - E_i}{\sqrt{E_i(1 - h_i)}}$$

where the leverage h_i is the i -th element on the diagonal of the hat matrix.

For the Poisson distribution, the residuals given by (6.5) and the chi-squared goodness of fit statistic are related by

$$X^2 = \sum_{i=1}^N r_i^2 = \sum_{i=1}^N \frac{(O_i - E_i)^2}{E_i}$$

which is the usual definition of the Pearson chi-squared statistic for contingency tables.

The deviance for a Poisson model has already been derived in Section 4.8.3. It can be written in the form

$$D = 2 \sum_{i=1}^N \left[O_i \log \left(\frac{O_i}{E_i} \right) - (O_i - E_i) \right] \quad (6.6)$$

However, for most models we have $\sum O_i = \sum E_i$, so that the deviance simplifies to

$$D = 2 \sum_{i=1}^N \left[O_i \log \left(\frac{O_i}{E_i} \right) \right] \quad (6.7)$$

The **deviance residuals** are the components of D in (6.6), namely

$$d_i = \text{sign}(O_i - E_i) \sqrt{2 \left[O_i \log \left(\frac{O_i}{E_i} \right) - (O_i - E_i) \right]} \quad (6.8)$$

so that $D = \sum d_i^2$.

6.2.4 Goodness of fit

As in the Binomial case, the goodness of fit statistics X^2 and D are closely related.

Using a second order Taylor expansion it can be shown that

$$\begin{aligned} D &\approx 2 \sum_{i=1}^N \left[(O_i - E_i) + \frac{1}{2} \frac{(O_i - E_i)^2}{E_i} - (O_i - E_i) \right] \\ &= \sum_{i=1}^N \left[\frac{(O_i - E_i)^2}{E_i} \right] = X^2 \end{aligned}$$

If there is no over-dispersion (see Section 6.8), the statistics D and X^2 can be used directly as measures of goodness of fit, since they can be calculated from the data and the fitted model and do not involve any nuisance parameters (like σ^2 for the Normal distribution). They can be compared to the Chi-square distribution with $N - p$ degrees of freedom, where p is the number of parameters estimated. The Chi-square distribution is likely to be a better approximation to the sampling distribution of X^2 than of D .

6.2.5 Example: Poisson regression

The data in Table 6.1 come from a famous study conducted by Sir Richard Doll and colleagues. In 1951, all British doctors were sent a brief questionnaire about whether

they smoked tobacco. Thereafter, from 1951 onwards, information about their deaths has been collected. Table 6.1 shows the number of deaths from coronary heart disease among male doctors 10 years after the survey. The table also shows the total number of person-years of observations at the time of the analysis (Breslow and Day, 1987). The questions of interest are:

1. Is the death rate higher for smokers than for non-smokers?
2. If so, by how much?
3. Is the effect of smoking related to age?

Table 6.1: Deaths from Coronary Heart Disease

Age Group	Smokers		Non-Smokers	
	Deaths	Person-years	Deaths	Person-years
35-44	32	52407	2	18790
45-54	104	43248	12	10673
55-64	206	28612	28	5710
65-74	186	12663	28	2585
75-84	102	5217	31	1462

We fit the following model to the data:

$$\begin{aligned}
 \log[E(Y_i)] &= \log(\mu_i) \\
 &= \log(n_i) + \beta_1 + \text{smoke}_i\beta_2 + \text{agecat}_i\beta_3 + \text{agesqu}_i\beta_4 + \text{smokeage}_i\beta_5
 \end{aligned} \tag{6.9}$$

Here Y_i is the number of deaths; n_i the number of personyears; smoke_i is 1 for smokers and 0 for non-smokers (smoking status); agecat_i runs from 1 to 5 for the 5 age groups in Table 6.1, to model age dependency of the death rate; agesqu_i is the square of agecat_i , used to model a nonlinear increase in death rate as a function of age; and smokeage_i is the interaction of smoking status and age, used to model a differential effect of age depending on smoking status.

The three components of the generalized linear model are then as follows:

1. **Error structure / Distribution:** Independent response variables

$$Y_i \sim \text{Poisson}(\mu_i).$$

2. **Linear Predictor:** Parameters β and explanatory variables X such that

$$\eta_i = x_i' \beta \text{ as specified in Equation (6.9) above.}$$

3. **Link Function:** The link function g is the natural logarithm, namely

$$g(\mu_i) = \log(\mu_i) = \log(n_i) + x_i' \beta$$

The SAS code to fit the Poisson regression model (6.9) is as follows:

```
PROC GENMOD DATA=smokers;
  TITLE 'Poisson Regression:  Smokers versus Non-Smokers';
  MODEL deaths = smoke agecat agesqu smoke*agecat
    / OFFSET=logyears DIST=poisson LINK=log TYPE3 PREDICTED RESIDUALS
RUN;
```

Note the specification of the offset variable, namely $\log(\text{personyears})$, as an option in the **MODEL** statement. The **PREDICTED** and **RESIDUALS** options lead SAS to print out predicted (fitted) values and both the Pearson and deviance residuals, both unstandardized and standardized.

The SAS output is included with Appendix 6.1.

The model seems to fit well, based on a deviance value of $D = 1.635$ on 5 degrees of freedom. Furthermore, all fitted values are close to the observed values, and consequently all residuals are small (standardized residuals all smaller than 0.9).

The Wald statistics for all model parameters are highly significant. The parameter estimate associated with smoking status is $\hat{\beta}_2 = 1.441$, which corresponds to a rate ratio of $RR = e^{1.441} = 4.2$. Therefore, smokers have a more than 4 times higher death rate due to coronary heart disease than non-smokers. However, as age increases, the effect of smoking becomes smaller, since the regression coefficient for the interaction of age and smoking is negative. In the highest age group, the rate ratio is approximately 1 (the rate ratio in that age group is $RR = e^{(1.44 - 5 \cdot 0.31)} = 0.90$, which can be confirmed from the raw data in Table 6.1.)

6.3 Contingency Tables

Before specifying log-linear models for frequency data summarized as contingency tables, it is important to consider how the design of a study may determine constraints on the data. The study design also affects the choice of probability model to describe the data. These issues are illustrated in the following examples.

6.3.1 Example: Cross-sectional study

Table 6.2 summarizes the data from a cross-sectional study of patients with malignant melanoma (skin cancer). For a sample of $n = 400$ patients, the site of the tumor and its type were recorded.

Table 6.2: Malignant Melanoma by Tumor Type and Site

Tumor type	Tumor Site			
	Head&Neck	Trunk	Extremities	Total
Hutchinson's melanomic freckle	22	2	10	34
Superficial spreading melanoma	16	54	115	185
Nodular	19	33	73	125
Indeterminate	11	17	28	56
Total	68	106	226	400

The question of interest is whether there is an association between tumor type and site. In Appendix 6.2 the data is displayed including row and column percentages.

Let Y_{jk} denote the frequency for the (j, k) -th cell of the contingency table, with $j = 1, \dots, J$ and $k = 1, \dots, K$. In the example there are $J = 4$ rows and $K = 3$ columns, and we have the constraint that $\sum_{j=1}^J \sum_{k=1}^K Y_{jk} = 400$ is fixed by the design of the study. If the Y_{jk} are independent random variables with Poisson distributions with parameters $E(Y_{jk}) = \mu_{jk}$, then their sum has Poisson distribution with parameter $E(n) = \mu = \sum_{j=1}^J \sum_{k=1}^K \mu_{jk}$. Therefore the joint distribution of the Y_{jk} 's, conditional

on their sum n , is the multinomial distribution

$$f(y|n) = n! \prod_{j=1}^J \prod_{k=1}^K \frac{\theta_{jk}^{y_{jk}}}{y_{jk}!}$$

where $\theta_{jk} = \mu_{jk}/\mu$. The sum of the terms θ_{jk} is one because $\mu = \sum_{j=1}^J \sum_{k=1}^K \mu_{jk}$, and $0 < \theta_{jk} < 1$. Therefore the θ_{jk} can be interpreted as the probability that a subject falls into the (j, k) -th cell of the table. Furthermore, the expected value of Y_{jk} is

$$E(Y_{jk}) = \mu_{jk} = n\theta_{jk}$$

The usual link function for a Poisson model gives

$$\log(\mu_{jk}) = \log(n) + \log(\theta_{jk})$$

which looks like Equation (6.3) except that now the offset $\log(n)$ is the same for all observations.

6.3.2 Example: Clinical trial / prospective study

In a prospective study of a new vaccine for influenza, patients were randomly allocated to either placebo or the new vaccine. The responses were titre levels of antibody found in the blood six weeks after vaccination, categorized as small, moderate or large. The cell frequencies in the rows are constrained to add up to the number of subjects in each treatment group (38 and 35, respectively). The question of interest is whether the pattern of responses is the same for each treatment group.

Table 6.3: Influenza Vaccine Trial

Treatment	Response			Total
	Small	Moderate	large	
Placebo	25	8	5	38
Vaccine	6	18	11	35

In this example the row totals are fixed. Thus the joint probability distribution for each row is multinomial

$$f(y_{j1}, \dots, y_{jK} \mid y_{j+}) = y_{j+}! \prod_{k=1}^K \frac{\theta_{jk}^{y_{jk}}}{y_{jk}!}$$

where $y_{j+} = \sum_{k=1}^K y_{jk}$ is the row total and $\sum_{k=1}^K \theta_{jk} = 1$. So the joint probability distribution for all the cells in the table is the **product multinomial distribution**

$$f(y \mid y_{1+}, \dots, y_{J+}) = \prod_{j=1}^J y_{j+}! \prod_{k=1}^K \frac{\theta_{jk}^{y_{jk}}}{y_{jk}!}$$

where $\sum_{k=1}^K \theta_{jk} = 1$ for each row. In this case we have $E(Y_{jk}) = y_{j+}\theta_{jk}$ so that

$$\log[E(Y_{jk})] = \log(\mu_{jk}) = \log(y_{j+}) + \log(\theta_{jk})$$

If the response pattern for both treatments was the same, then $\theta_{jk} = \theta_{.k}$ for $k = 1, \dots, K$.

6.3.3 Example: Retrospective study

In a case-control (retrospective) study, a group of ulcer patients was compared to a group of control patients without peptic ulcers, but who were similar to the ulcer patients with respect to age, sex and socio-economic status. The ulcer patients were classified according to the site of the ulcer (gastric or duodenal), and aspirin use was ascertained for all patients. The results are shown in Table 6.4.

Table 6.4: Gastric Ulcers and Aspirin Use

Ulcer Type	Group	Aspirin Use		
		Non-user	User	Total
Gastric	Controls	62	6	68
	Cases	39	25	64
Duodenal	Controls	53	8	61
	Cases	49	8	57

This is a stratified 2×2 table, namely a table of group $\times$ aspirin-use, stratified by ulcer type. Some questions of interest are:

1. Is gastric ulcer associated with aspirin use?
2. Is duodenal ulcer associated with aspirin use?

3. Is any association with aspirin use the same for the two ulcer types?

In this example the numbers of patients in each of the control and cases groups, that is, all 4 row totals in Table 6.4, were fixed.

Let $j = 1, 2$ denote the controls or cases, $k = 1, 2$ denote gastric versus duodenal ulcers, and $l = 1, 2$ denote aspirin non-use and aspirin use. In general, let Y_{jkl} denote the frequency of observations in category (j, k, l) with $j = 1, \dots, J$, $k = 1, \dots, K$ and $l = 1, \dots, L$. If the marginal totals y_{jk+} are fixed, the joint probability distribution for all the cells in the table is

$$f(y \mid y_{11+}, \dots, y_{JK+}) = \prod_{j=1}^J \prod_{k=1}^K y_{jk+}! \prod_{l=1}^L \frac{\theta_{jkl}^{y_{jkl}}}{y_{jkl}!}$$

where $\sum_{l=1}^L \theta_{jkl} = 1$ for $j = 1, \dots, J$ and $k = 1, \dots, K$. This is another form of the **product multinomial distribution**. In this case, $E(Y_{jkl}) = \mu_{jkl} = y_{jk+} \theta_{jkl}$ so that

$$\log[E(Y_{jkl})] = \log(\mu_{jkl}) = \log(y_{jk+}) + \log(\theta_{jkl})$$

6.4 Probability Models for Contingency Tables

The examples in Sections 6.3.1 to 6.3.3 illustrate the main probability models for contingency table data. In general, let the vector y denote the frequencies Y_i in N cells of a cross-classified table.

6.4.1 Poisson model

If there were no constraints on the Y_i 's they can be modelled as independent random variables with parameters $E(Y_i) = \mu_i$ and joint probability distribution

$$f(y \mid \mu) = \prod_{i=1}^N \mu_i^{y_i} e^{-\mu_i} y_i!$$

which is a product Poisson distribution.

6.4.2 Multinomial model

If the only constraint is that the sum of the Y_i 's is n , then the multinomial distribution may be used

$$f(y \mid \mu, n) = n! \prod_{i=1}^N \frac{\theta_i^{y_i}}{y_i!}$$

where $\sum_{i=1}^N \theta_i = 1$ and $\sum_{i=1}^N y_i = n$. In this case $E(Y_i) = n\theta_i$.

For a two-dimensional contingency table like Table 6.2, if j and k denote the rows and columns, then the most commonly considered hypothesis is that the row and column variables are independent so that

$$\theta_{jk} = \theta_{j+}\theta_{+k}$$

where θ_{j+} and θ_{+k} denote the marginal probabilities. The hypothesis of independence can be tested by comparing the fit of two linear models for the logarithm of $\mu_{jk} = E(Y_{jk})$, namely by comparing the “saturated” model

$$\log(\mu_{jk}) = \log(n) + \log(\theta_{jk})$$

with the model assuming independence of the row and column variables, namely

$$\log(\mu_{jk}) = \log(n) + \log(\theta_{j+}) + \log(\theta_{+k})$$

6.4.3 Product Multinomial models

If there are more fixed marginal totals than just the overall total n , then the appropriate products of multinomial distributions can be used to model the data.

For example, for a three-dimensional table with J rows, K columns and L strata, if the row totals are fixed for each stratum the joint probability for the Y_{jkl} 's is

$$f(y \mid y_{j+l}; j = 1, \dots, J, l = 1, \dots, L) = \prod_{j=1}^J \prod_{l=1}^L y_{j+l}! \prod_{k=1}^K \frac{\theta_{jkl}^{y_{jkl}}}{y_{jkl}!}$$

where $\sum_{k=1}^K \theta_{jkl} = 1$ for each combination of j and l . In this case $E(Y_{jkl}) = y_{j+l}\theta_{jkl}$.

If only the stratum totals are fixed, then

$$f(y \mid y_{++l}, l = 1, \dots, L) = \prod_{l=1}^L y_{++l}! \prod_{j=1}^J \prod_{k=1}^K \frac{\theta_{jkl}^{y_{jkl}}}{y_{jkl}!}$$

with $\sum_{j=1}^J \sum_{k=1}^K \theta_{jkl} = 1$ for $l = 1, \dots, L$. Also $E(Y_{jkl}) = y_{++l}\theta_{jkl}$.

6.5 Log-Linear Models

All probability models given in Section 6.4 are based on the Poisson distribution and in all cases $E(Y_i)$ can be written as the product of parameters and of “offsets”. Thus the natural link function for the Poisson distribution, the logarithmic function, yields a linear component

$$\log[E(Y_i)] = \text{constant} + x'_i\beta$$

The term **log-linear model** is used to describe all these types of generalized linear models.

For the melanoma example of Section 6.3.1 (cross-sectional study), if there is no association between site of tumor and type of tumor, so that these two variables are independent, their joint probability is the product of the marginal probabilities

$$\theta_{jk} = \theta_{j+}\theta_{+k}, \quad j = 1, \dots, J \text{ and } k = 1, \dots, K$$

The hypothesis of independence can be tested by comparing the additive model (on the logarithmic scale)

$$\log[E(Y_{jk})] = \log(\mu_{jk}) = \log(n) + \log(\theta_{j+}) + \log(\theta_{+k}) \quad (6.10)$$

with the “full” or “saturated” model

$$\log[E(Y_{jk})] = \log(\mu_{jk}) = \log(n) + \log(\theta_{jk}) \quad (6.11)$$

This is analogous to a two-way analysis of variance (ANOVA) for a two-factor experiment: Model (6.10) is the main effects model, while Model (6.11) fits the two main effects and their interaction. In ANOVA notation, therefore, Models (6.10) and (6.11) could alternatively be written as

$$\log[E(Y_{jk})] = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} \quad (6.12)$$

for the full model, and

$$\log[E(Y_{jk})] = \mu + \alpha_j + \beta_k \quad (6.13)$$

for the additive model. Since the term $\log(n)$ has to be in all models, the minimal model is

$$\log[E(Y_{jk})] = \mu$$

For the flu vaccine trial of Section 6.3.2 (prospective study)

$$\log[E(Y_{jk})] = \log(y_{j+}) + \log(\theta_{jk}) \quad (6.14)$$

if the distribution of responses described by the θ_{jk} differs for the J groups. Alternatively

$$\log[E(Y_{jk})] = \log(y_{j+}) + \log(\theta_{\cdot k}) \quad (6.15)$$

if it is the same for all groups. So the hypothesis of **homogeneity** of the response distributions can be tested by comparing the “full” or “saturated” model

$$\log[E(Y_{jk})] = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} \quad (6.16)$$

corresponding to (6.14) with the additive model

$$\log[E(Y_{jk})] = \mu + \alpha_j + \beta_k \quad (6.17)$$

corresponding to (6.15). The minimal model for this data is

$$\log[E(Y_{jk})] = \mu + \alpha_j$$

because the row totals, corresponding to the subscript j , are fixed (and known!) by the design of the study.

More generally, the specification of the linear components for log-linear models bears many resemblances to the specification of ANOVA models. The models are **hierarchical**, meaning that if a higher order (interaction) term is included in the model then all the related lower order terms must also be included in the model. Thus, if the two-factor interaction $(\alpha\beta)_{jk}$ is included in the model, then so are the two associated main effects α_j and β_k , and the “intercept” μ . Similarly, if three-factor interactions $(\alpha\beta\gamma)_{jkl}$ are included then so are the two-factor interactions $(\alpha\beta)_{jk}$, $(\alpha\gamma)_{jl}$ and $(\beta\gamma)_{kl}$, all three main effects, and the intercept.

For contingency tables, the main questions almost always relate to associations between variables. Therefore, in log-linear models, the terms of primary interest are the interactions involving two or more variables.

6.6 Inference for Log-Linear Models

Although three types of probability distributions are used to describe contingency table data (Poisson, multinomial, product multinomial; see Section 6.4), Birch (1963) showed that for any log-linear model the maximum likelihood estimators are the same for all these distributions provided that the parameters which correspond to the fixed (and known) marginal totals are always included in the model. This means that for the purpose of estimation the Poisson distribution can always be assumed. As the Poisson distribution belongs to the exponential family and the parameter constraints can be incorporated in the linear component, all the standard methods for generalized linear models can be used.

The adequacy of a model can be assessed using the goodness of fit statistics X^2 (Pearson chi-square statistic) or D (deviance) summarized in Section 6.2 for Poisson regression. Hypothesis tests can be conducted by comparing the goodness of fit statistic of a larger or general model (corresponding to the alternative hypothesis) to the goodness of fit statistic of a simpler nested model (corresponding to the null hypothesis). These methods are illustrated in the following examples.

6.7 Examples: Log-Linear Models

6.7.1 Cross-sectional study of malignant melanoma

We are using SAS PROC GENMOD to fit log-linear models (6.12) and (6.13) to the data of Table 6.2. The three components of the generalized linear model are as follows:

1. **Error structure / Distribution:** Response variables $Y_{jk} \sim \text{Poisson}(n\theta_{jk})$.
2. **Linear Predictor:** Parameters and explanatory variables such that $\eta_{jk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$ as specified in Equation (6.12) above (for illustration we also fit the simpler additive model with only the main effects).
3. **Link Function:** The link function g is the natural logarithm, namely $g(\mu_{jk}) = \log(\mu_{jk})$

The SAS code to fit the log-linear model (6.13) is as follows:

```
PROC GENMOD DATA=melanoma;  
    TITLE 'Log-Linear Model: Cross-Sectional Study';  
    CLASS type site;  
    MODEL count = type site type*site / DIST=poisson LINK=log TYPE3 PREDICTED;  
RUN;
```

The SAS MODEL statement fitting the main effects model (assuming independence between tumor type and tumor site) is

```
MODEL count = type site / DIST=poisson LINK=log TYPE3 PREDICTED;
```

The output for both models is included with Appendix 6.2.

Looking at the output for Model (6.12), which is the full or saturated model, the LR statistic for the type by site interaction term is 51.80 with 6 degrees of freedom ($P < 0.0001$). Thus the null hypothesis of independence between tumor site and tumor type can be rejected. The LR statistic is obtained by the specification of the TYPE3 option in the MODEL statement. Specifying the WALD option, one obtains the Wald statistic for the interaction term, instead of the LR statistic. In fact the LR statistic for the interaction term type $\times$ site is the difference of the deviance D of the full/saturated model, which by definition is zero, and the deviance of the main effects model (6.13): note the deviance for the main effects model in the SAS output: 51.795. Thus, when using the TYPE3 option, it is in fact not necessary to fit both model (6.13) and (6.12), calculate the respective deviances and subtract them from each other, in order to obtain the LR test statistic for the interaction term.

Another test statistic (in addition to the LR and Wald tests) for the interaction term is the Pearson chi-square statistic X^2 : its value is 65.81 with 6 degrees of freedom, also highly significant.

Furthermore, in the output for the saturated model we note that the observed and predicted counts (produced by the PREDICTED option in the MODEL statement) are identical, again by definition. This corresponds to the fact that the deviance and Pearson chi-square statistics for the saturated model are zero.

Note: For this relatively simple example (simple two-way contingency table) the conventional analysis of a two-way table (as could be carried out by the SAS procedure FREQ; see lecture notes for “Categorical Data Analysis”) produces exactly the same results as the log-linear analysis above. The great advantage of log-linear modeling is that it provides a method for analyzing more complicated cross-tabulated data, as illustrated in the next example.

6.7.2 Case-control study of ulcers and aspirin use

Even a brief look at Table 6.4 suggests that aspirin use may be a risk factor for gastric ulcers but not for duodenal ulcers. In a formal log-linear analysis we initially fit the full or saturated model for this data, namely the three main effects case-control status, ulcer type, and aspirin use, plus the three two-way interactions between those three effects, plus the three-way interaction. The three components of the generalized linear model are as follows:

1. **Error structure / Distribution:** Response variables $Y_{jkl} \sim \text{Poisson}(y_{jk+} \cdot \theta_{jkl})$, where the indices j and k indicate case-control status and ulcer type, respectively.
2. **Linear Predictor:** Parameters and explanatory variables such that the linear predictor represents a fully saturated three-way layout:

$$\eta_{jkl} = \mu + \alpha_j + \beta_k + \gamma_l + (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + (\beta\gamma)_{kl} + (\alpha\beta\gamma)_{jkl}$$

3. **Link Function:** The link function g is the natural logarithm, namely $g(\mu_{jkl}) = \log(\mu_{jkl})$.

The SAS code to fit the saturated log-linear model as follows:

```
PROC GENMOD DATA=ulcer;
  TITLE 'Log-Linear Model: Stratified Case-Control Study';
  CLASS casecontrol ulcertype aspirinuse;
  MODEL count = casecontrol|ulcertype|aspirinuse
              / DIST=poisson LINK=log TYPE3 PREDICTED;
RUN;
```

For illustrative purposes, we also fit a simpler model with the three-way interaction removed. The corresponding `MODEL` statement is as follows

```
MODEL count = casecontrol|ulcertype
          aspirinuse aspirinuse*ulcertype aspirinuse*casecontrol
          / DIST=poisson LINK=log TYPE3 PREDICTED;
```

Note that in all models the main effects `casecontrol` and `ulcertype`, and their interaction, must be fitted, because these correspond to the fixed (and known) marginal totals by the design of the study.

The output is included with Appendix 6.3.

Generally, the aim of the analysis is to determine whether there is an association between case control status and aspirin use. That is, in statistical terms, the question is whether the interaction `casecontrol`×`aspirinuse` is significant. We have, however, a third factor: ulcer type, and it is possible that the association between case control status and aspirin use depends on ulcer type: in statistical terms, the interaction `casecontrol`×`ulcertype`×`aspirinuse` could be significant. The latter is in fact suggested by the observation that, looking at the table of frequencies, aspirin use seems to be a risk factor for gastric ulcers but not for duodenal ulcers.

In the output for the saturated model we note that the LR statistic for the three-way interaction, `casecontrol`×`ulcertype`×`aspirinuse`, is 6.28 with 1 degree of freedom ($P=0.0122$). Thus the three-way interaction is indeed significant: we can conclude that the association between case control status and aspirin use depends on ulcer type. From here, the analysis could proceed by analyzing the data for the two strata (ulcer type) separately, namely by investigating further the association between aspirin use and case control status, within each ulcer type.

However, for illustrative purposes we proceed as if the interaction had not been significant. In this case we would drop the three-way interaction, and fit the reduced model without the three-way interaction. Then, in the reduced model, we investigate the association between case control status and aspirin use: specifically, we investigate whether the interaction `casecontrol`×`apsirinuse` is significant.

In the output for the reduced model we note that the LR statistic for the two-way interaction, $\text{casecontrol} \times \text{aspirinuse}$, is 11.41 with 1 degree of freedom ($P=0.0007$), which is highly significant. We can conclude there is a significant association between aspirin use and case-control status, which allows us to conclude, given the observed and fitted frequencies, that aspirin use is associated with increased risk of ulcer. The rate ratio is 3.13 with a 95% CI of 1.57 to 6.25 which can be read (on the log scale) from the output page that lists the parameter estimates.

6.8 Over-Dispersion

As with the Binomial case (Section 5.7), over-dispersion for Poisson models can be handled either through a model assuming a Negative Binomial distribution, or through estimation of and subsequent scaling by a dispersion parameter.

6.8.1 Example: Over-dispersion for count data

A number of larvae were diluted in an amount of water, and subsequently 20 small aliquots from the water were taken in five batches of four aliquots each, and the number of larvae in each aliquot were counted. The aim of the study was to find out how reproducible (similar) the number of larvae in each aliquot were. If the numbers were very similar, in practical applications it would not be necessary to count the number of larvae in each aliquot used, but only in small sample of aliquots. – The experiment was carried out twice (count 1 and count 2), but here we only consider the data for count 1.

The data are presented in Table 6.5:

Table 6.5: Larvae Counted in Aliquots

Aliquot	Batch	Count 1	Count 2
1	1	1288	653
2	1	1325	656
3	1	1365	692
4	1	1283	740
5	2	1219	540
6	2	1207	740
7	2	1239	749
8	2	1219	762
9	3	1353	766
10	3	1435	771
11	3	1351	774
12	3	1305	783
13	4	1234	798
14	4	1152	798
15	4	1213	816
16	4	1108	829
17	5	1019	873
18	5	1008	900
19	5	1108	927
20	5	920	1094

We begin by fitting a Poisson model to these count data, with Batch as a categorical variable. Since the aliquots were each of the same size, no offset variable is required (if aliquots of different sizes had been taken, aliquot size would have to be fitted as an offset variable).

The three components of the generalized linear model are as follows:

1. **Error structure / Distribution:** Response variables $Y_{ij} \sim \text{Poisson}(\mu_{ij})$, where the indices i and j indicate aliquot number and batch number, respectively.

2. **Linear Predictor:** Parameters and explanatory variables such that the linear predictor represents a one-way layout: $\eta_{ij} = \mu + \alpha_j$
3. **Link Function:** The link function g is the natural logarithm, namely $g(\mu_{ij}) = \log(\mu_{ij})$.

The SAS code to fit this model as follows:

```
PROC GENMOD DATA=larvae;
  TITLE 'Poisson Model: Over-Dispersion for Larvae Data';
  CLASS batch;
  MODEL count1 = batch
          / DIST=poisson LINK=log TYPE3 PREDICTED RESIDUALS SCALE=p;
RUN;
```

The output is included with Appendix 6.4.

We notice on page 1 of the output that the both the Pearson chi-square statistic and the deviance are significantly larger than their degrees of freedom, by a factor of about 2.4. Thus we clearly have a case of over-dispersion. To account for over-dispersion in hypothesis tests and confidence interval calculations, we have specified in the **MODEL** statement the **SCALE=pearson** option, which causes SAS to estimate the dispersion parameter based on the Pearson chi-square statistic and make the relevant adjustment to test statistics and interval estimates (in Equations (5.9) and (5.10) the unknown value of σ^2 is replaced by its estimate from (5.11)).

As in the binomial case, the estimate of the scale (dispersion) parameter on page two of the output (labeled “Scale”) is for σ , thus it is equal to the square root of the ratio of the Pearson chi-square statistic over its degrees of freedom (the latter labeled “Value/DF” by SAS).

The factor Batch is significant, so that we can conclude that there is significant batch-to-batch variation, but even accounting for the variation due to batch in the model the data is still significantly over-dispersed relative to the Poisson distribution.

Chapter 7

Data with Constant Coefficient of Variation

7.1 Introduction

In the classical linear model with Normally distributed data the variance of the observations, σ^2 , is assumed constant over the entire range of parameter values β . In Chapter 6 we considered models for **count** data (e.g. with a Poisson distribution) where $\text{Var}(Y) \propto \text{E}(Y)$. However, in practice one commonly has data also in the form of **continuous** measurements where the variance increases with the mean. In the present chapter we assume that the variance of the data is of the following form:

$$\text{Var}(Y) = \sigma^2 [\text{E}(Y)]^2 = \sigma^2 \mu^2 \quad (7.1)$$

for some constant σ^2 . Equation (7.1) implies that

$$\text{CV}(Y) = \frac{\sqrt{\text{Var}(Y)}}{\text{E}(Y)} = \frac{\sigma\mu}{\mu} = \sigma$$

so that the coefficient of variation $\text{CV}(Y)$ of Y is constant over all μ . (Note that here σ is the coefficient of variation of the random variable, not its standard deviation.)

7.1.1 Option 1: Logarithmic transformation of data

If a random variable Y has variance function as in (7.1) then, for small σ , the variance stabilizing transformation is the logarithm, since the log-transformed variable $\log(Y)$

has approximate variance

$$\begin{aligned}\text{Var}[\log(Y)] &\approx \left[\frac{\partial \log(y)}{\partial y} \Big|_{y=\mu} \right]^2 \cdot \text{Var}(Y) \\ &= \frac{1}{\mu^2} \cdot \sigma^2 \mu^2 \\ &= \sigma^2\end{aligned}$$

Furthermore, if the systematic part of the model is multiplicative on the original scale, then the model is additive (linear) on the logarithmic scale, that is, we have

$$\eta_i = \log[E(Y_i)] = x_i' \beta$$

Thus, in this case, a linear model can be fitted on the logarithmic scale and parameter estimates can be obtained using ordinary least squares.

7.1.2 Option 2: Gamma model for data

Alternatively to logarithmic transformation of the data we might decide to analyse the data on the original scale. In this case, we assume a logarithmic link function between μ and the linear predictor, that is, we have

$$\mu = E(Y) = e^{x_i' \beta}$$

Furthermore, as in (7.1), we assume a quadratic variance function (that is, the variance of Y is proportional to the square of μ) to describe the relationship between mean and variance. With these assumptions about the link function and the variance function, iteratively weighted least squares can be used to obtain parameter estimates for β using the algorithm described in Section 3.2. This method of estimation is equivalent to assuming that Y has the Gamma distribution with constant index $\nu = 1/\sigma^2$ independent of the mean.

Thus, when it is reasonable to assume that the data at hand have constant coefficient of variation, one modeling option is to fit a generalized linear model assuming a Gamma distribution.

7.2 Gamma Distribution

7.2.1 Mean, variance, coefficient of variation

As we have seen in Section 2.2.4, the pdf of the Gamma distribution is

$$f(y; \mu) = \frac{1}{\Gamma(\nu)y} \left(\frac{\nu y}{\mu} \right)^\nu \exp \left(-\frac{\nu y}{\mu} \right) \quad (7.2)$$

which can be written as

$$f(y; \mu) = \exp \left[-\nu y \cdot \frac{1}{\mu} - \nu \log(\mu) + \nu [\log(y) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y) \right]$$

so that

$$\begin{aligned} a(y) &= -\nu y \\ b(\mu) &= \frac{1}{\mu} \\ c(\mu) &= -\nu \log(\mu) \\ d(y) &= \nu [\log(y) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y) \end{aligned}$$

Written in the above form the distribution is not in canonical form since $a(y) = -\nu y$.

The natural parameter of the distribution is $b(\mu) = 1/\mu$.

Using expressions (2.13) and (2.15) for the mean and variance of $a(Y)$ respectively, we obtain expressions for the mean and variance of a random variable with Gamma distribution:

$$\begin{aligned} E[a(Y)] = -\nu E(Y) &= -\frac{c'}{b'} \\ &= -\frac{-\nu}{\mu} \cdot \frac{-\mu^2}{1} \\ &= -\nu\mu \end{aligned}$$

so that

$$E(Y) = \mu \quad (7.3)$$

Similarly,

$$\text{Var}[a(Y)] = \nu^2 \text{Var}(Y) = \frac{b''c' - c''b'}{(b')^2}$$

$$\begin{aligned}
&= \left(\frac{2}{\mu^3} \cdot \frac{-\nu}{\mu} - \frac{\nu}{\mu^2} \cdot \frac{-1}{\mu^2} \right) \cdot \left(-\frac{1}{\mu^2} \right)^{-3} \\
&= \left(\frac{-2\nu + \nu}{\mu^4} \right) \cdot (-\mu^6) \\
&= \nu\mu^2
\end{aligned}$$

so that

$$\text{Var}(Y) = \frac{\mu^2}{\nu} \quad (7.4)$$

The coefficient of variation of Y is therefore given by

$$\begin{aligned}
\text{CV}(Y) &= \frac{\sqrt{\text{Var}(Y)}}{\text{E}(Y)} \\
&= \frac{\mu}{\sqrt{\nu}\mu} \\
&= \frac{1}{\sqrt{\nu}}
\end{aligned}$$

Thus, when modeling a sample of observations from the Gamma distribution for which the parameter ν is constant for all observations, the coefficient of variation of the data is indeed constant (independent of μ).

7.2.2 Deviance

If $Y_i \sim \Gamma(\mu_i, \nu)$ are independent, the log-likelihood function is

$$l(\mu; y) = \sum_{i=1}^N \left\{ -\nu \left[\frac{y_i}{\mu_i} + \log(\mu_i) \right] + \nu [\log(y_i) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y_i) \right\}$$

For a saturated model, all the μ_i are potentially different and the maximum likelihood estimates of the μ_i are $\hat{\mu}_i = y_i$. Thus the maximum value of the log-likelihood function is

$$\begin{aligned}
l(\mu_{\max}; y) &= \sum_{i=1}^N \left\{ -\nu \left[\frac{y_i}{y_i} + \log(y_i) \right] + \nu [\log(y_i) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y_i) \right\} \\
&= \sum_{i=1}^N \left\{ -\nu [1 + \log(y_i)] + \nu [\log(y_i) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y_i) \right\}
\end{aligned}$$

Under some model with $p < N$ parameters, we denote the maximum likelihood estimates of the μ_i by $\hat{\mu}_i$, which are equal to the fitted values $\hat{y}_i$. The value of the

log-likelihood function is then

$$l(\hat{\mu}; y) = \sum_{i=1}^N \left\{ -\nu \left[\frac{y_i}{\hat{y}_i} + \log(\hat{y}_i) \right] + \nu [\log(y_i) + \log(\nu)] - \log[\Gamma(\nu)] - \log(y_i) \right\}$$

The deviance is therefore given by

$$\begin{aligned} D &= 2[l(\mu_{\max}; y) - l(\hat{\mu}; y)] \\ &= 2 \left\{ \sum_{i=1}^N -\nu [1 + \log(y_i)] - \sum_{i=1}^N -\nu \left[\frac{y_i}{\hat{y}_i} + \log(\hat{y}_i) \right] \right\} \\ &= -2\nu \left\{ \sum_{i=1}^N [1 + \log(y_i)] - \sum_{i=1}^N \left[\frac{y_i}{\hat{y}_i} + \log(\hat{y}_i) \right] \right\} \\ &= -2\nu \left\{ \sum_{i=1}^N \log \left(\frac{y_i}{\hat{y}_i} \right) - \frac{y_i - \hat{y}_i}{\hat{y}_i} \right\} \end{aligned}$$

Note that the deviance statistic is defined only if all observations are strictly positive, but see McCullagh and Nelder (1989, p. 290) on how to handle the case when some of the y_i are zero.

7.3 Link Functions

7.3.1 Canonical link: inverse link

As we have seen in Section 7.2.1, the natural parameter of the Gamma distribution is $\theta = b(\mu) = 1/\mu$. Thus the canonical link function is the inverse link

$$\mu^{-1} = \eta = X\beta$$

Unlike the canonical links for the Binomial and Poisson distributions (the logit and the log, respectively), the inverse link (reciprocal transformation) does not map the range of $\mu > 0$ onto the whole real line. Thus the requirement that $\eta > 0$ implies restrictions on the β 's in any linear model so that negative values of $\hat{\mu}$ are avoided.

An example of the canonical link is given by the inverse polynomial response surfaces discussed by Nelder (1966). The most simple case, inverse linear response, is given by

$$\eta = \beta_0 + \frac{1}{x}\beta_1, \quad \text{where } x > 0 \quad (7.5)$$

In plant density experiments it is commonly observed that the yield per plant varies inversely with plant density x , so that the mean yield per plant has the form $1/(\beta_1 + x\beta_0)$. The yield per unit area is then given by

$$\eta^{-1} = \mu = \frac{x}{x\beta_0 + \beta_1} \quad (7.6)$$

Model (7.6) has an asymptote at $1/\beta_0$. Taking the inverse of (7.6) we obtain the linear model (7.5).

Inclusion in (7.5) of a linear term in x gives

$$\eta = \beta_0 + \frac{1}{x}\beta_1 + x\gamma_1$$

which is called the inverse quadratic.

These models (surfaces) can be extended to include more than one covariate and to include cross-terms in $1/(x_1x_2)$, x_1/x_2 and so on. For positive values of the parameters the surfaces have the desirable property that the ordinate η is everywhere positive and bounded.

7.3.2 Multiplicative models: log link

By combining the log link with terms linear in x and $1/x$ a large variety of response functions can be generated. These curves are sometime useful for describing response functions that have horizontal or vertical asymptotes, or functions that have turning points but are clearly asymmetric about that point.

$$\log(\mu) = \eta = \beta_0 + x\beta_1 + \frac{1}{x}\beta_2 \quad (7.7)$$

7.4 Example

Hunt et al. (1945) published data on the clotting time of blood, giving clotting time in seconds (y) for normal plasma diluted to nine different percentage concentrations with prothrombin-free plasma (u); clotting was induced by two lots of thromboplastin. The data are presented in Table 7.1.

Table 7.1: Clotting Times of Blood

u	Clotting Time (y)	
	Lot 1	Lot 2
5	118	69
10	58	35
15	42	26
20	35	21
30	27	18
40	25	16
60	21	13
80	19	12
100	18	12

Initial plots of the data suggest a log scale for u to produce inverse linearity. Thus models with inverse link and covariate $x = \log(u)$ are fitted. Initially, the following ANCOVA model is tried out:

$$\mu_{ij}^{-1} = \eta_{ij} = \beta_0 + \alpha_j + x_i\beta_1 + x_i(\alpha\beta)_j \quad (7.8)$$

where $i = 1, \dots, 9$ is the index specifying the nine dilutions, $j = 1, 2$ the index specifying the lot number, and $x_i = \log(u_i)$. Thus Model (7.8) contains the categorical effect (factor) Lot, $x = \log(u)$ as covariate, and the interaction between Lot and the covariate.

The three components of the generalized linear model are as follows:

1. **Error structure / Distribution:** Independent response variables
 $Y_{ij} \sim \text{Gamma}(\mu_{ij}, \nu)$.
2. **Linear Predictor:** Parameters β and explanatory variables X such that
 $\eta_{ij} = x'_i\beta$ as specified in Equation (7.8) above.
3. **Link Function:** The link function g is the inverse, namely
 $g(\mu_{ij}) = \mu_{ij}^{-1} = x'_i\beta$

The SAS code to fit the Gamma ANCOVA Model (7.8) is as follows:

```
PROC GENMOD DATA=blood;  
  TITLE 'Gamma ANCOVA Model:  Clotting Times of Blood';  
  MODEL y = lot x lot*x  
          / DIST=gamma LINK=power(-1) TYPE3 PREDICTED RESIDUALS;  
RUN;
```

The output is included with Appendix 7.1.

The Gamma distribution (like the Normal distribution) has a scale parameter, which by default is estimated by maximum likelihood. Note that the value reported in the SAS printout is in fact the inverse of the dispersion parameter of the distribution.

We see that, clearly, all terms in the model are highly significant. The residuals suggest that the model fits reasonably well, with the possible exception of the observation for $u = 5$. When these two observations are omitted, the fit improves and the dispersion is greatly reduced.

Bibliography

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